


7.0.1 UGC NET CSE | June 2014 | Part 2 | Question: 46


Manager's salary details are hidden from the employee. This is called as

- | | |
|---------------------------------|-------------------------------|
| A. Conceptual level data hiding | B. Physical level data hiding |
| C. External level data hiding | D. Local level data hiding |

ugcnetcse-june2014-paper2 databases

Answer key

7.0.2 UGC NET CSE | June 2014 | Part 2 | Question: 50


Match the following :

List – I
List – II

- | | |
|------------------------|------------------------|
| a. DDL | i. LOCK TABLE |
| b. DML | ii. COMMIT |
| c. TCL | iii. Natural Differece |
| d. BINARY
Operation | iv. REVOKE |

Codes :

- | | |
|---------------------------|---------------------------|
| A. a-ii; b-i; c-iii; d-iv | B. a-i; b-ii; c-iv; d-iii |
| C. a-ii; b-iii; c-i; d-iv | D. a-iv; b-i; c-ii; d-iii |

ugcnetcse-june2014-paper2 databases

Answer key

7.0.3 UGC NET CSE | June 2012 | Part 2 | Question: 7


In multiuser database if two users wish to update the same record at the same time, they are prevented from doing so by

- | | | | |
|------------|-------------|------------------|----------------|
| A. Jamming | B. Password | C. Documentation | D. Record lock |
|------------|-------------|------------------|----------------|

ugcnetcse-june2012-paper2 databases

Answer key

7.0.4 UGC NET CSE | June 2012 | Part 3 | Question: 9


The problem that occurs when one transaction updates a database item and the transaction fails for some reason is

- | | |
|-----------------------------|-----------------------------|
| A. Temporary Select Problem | B. Temporary Modify Problem |
| C. Dirty Read Problem | D. None |

ugcnetcse-june2012-paper3 databases

Answer key

7.0.5 UGC NET CSE | December 2012 | Part 2 | Question: 32


The User Work Area (UWA) is a set of Program variables declared in the host program to communicate the contents of individual records between

- | | |
|-----------------------------|-------------------------------------|
| A. DBMS and the Host record | B. Host program and the Host record |
| C. Host program and DBMS | D. Host program and Host language |

ugcnetcse-dec2012-paper2 databases

Answer key

7.0.6 UGC NET CSE | June 2013 | Part 3 | Question: 47

Match the following:

- | | |
|-------------------------------|---------------------------|
| a. Foreign keys | i. Domain constraint |
| b. Private key | ii. Referential integrity |
| c. Event control action model | iii. Encryption |
| d. Data security | iv. Trigger |

- A. a-iii, b-ii, c-i, d-iv
C. a-iii, b-iv, c-i, d-ii

- B. a-ii, b-i, c-iv, d-iii
D. a-i, b-ii, c-iii, d-iv

ugcnetcse-june2013-paper3 databases

Answer key

7.0.7 UGC NET CSE | December 2013 | Part 2 | Question: 12

GO BOTTOM and SKIP-3 commands are given one after another in a database file of 30 records. It shifts the control to

- | | |
|-----------------|-----------------|
| A. 28-th record | B. 27-th record |
| C. 3-rd record | D. 4-th record |

ugcnetcse-dec2013-paper2 databases

Answer key

7.0.8 UGC NET CSE | December 2013 | Part 3 | Question: 60

Match the following :

List – I

- a. Secondary Index
- b. Non-procedural Query language
- c. Closure of set of attributes
- d. Natural JOIN

List – II

- i. Functional Dependency
- ii. B-tree
- iii. Relational Algebraic Operation
- iv. Domain Calculas

Codes :

- A. a-i, b-ii, c-iv, d-iii
C. a-i, b-iii, c-iv, d-ii

- B. a-ii, b-i, c-iv, d-iii
D. a-ii, b-iv, c-i, d-iii

ugcnetcse-dec2013-paper3 databases

Answer key

7.0.9 UGC NET CSE | December 2015 | Part 2 | Question: 14

Match the following database terms to their functions :

List – I

- (a) Normalization
- (b) Data Dictionary
- (c) Referential Integrity
- (d) External Schema

List – II

- (i) Enforces match of primary key to foreign key
- (ii) Reduces data redundancy in a database
- (iii) Defines view(s) of the database for particular user(s)
- (iv) Contains metadata describing database structure

Codes :

- A. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
C. (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

- B. (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
D. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

Answer key **7.0.10 UGC NET CSE | December 2015 | Part 2 | Question: 18**

Which of the following provides the best description of an entity type?

- A. A specific concrete object with a defined set of processes (e.g. Jatin with diabetes)
- B. A value given to a particular attribute (e.g. height - 230 cm)
- C. A thing that we wish to collect data about zero or more, possibly real world examples of it may exist
- D. A template for a group of things with the same set of characteristics that may exist in the real world

ugcnetcse-dec2015-paper2 databases

Answer key **7.0.11 UGC NET CSE | December 2015 | Part 2 | Question: 19**

A data which improves the performance and accessibility of the database are called:

- A. Indexes
- B. User Data
- C. Application Metadata
- D. Data Dictionary

ugcnetcse-dec2015-paper2 databases

Answer key **7.0.12 UGC NET CSE | July 2016 | Part 2 | Question: 17**

Which of the following statement(s) is/are FALSE in the context of Relational DBMS?

- I. Views in a database system are important because they with access control by allowing users to see only a particular subset of the data in the database
 - II. E-R diagrams are useful to logically model concepts
 - III. An update anomaly is when it is not possible to store information unless some other, unrelated information is stored as well
 - IV. SQL is a procedural language
- A. I and IV only B. III and IV only C. I, II and III only D. II, III and IV only

ugcnetcse-july2016-paper2 databases

Answer key **7.0.13 UGC NET CSE | July 2016 | Part 3 | Question: 10**

Which of the following statements is true?

 D_1 : The decomposition of the schema $R(A, B, C)$ into $R_1(A, B)$ and $R_2(A, C)$ is always lossless D_2 : The decomposition of the schema $R(A, B, C, D, E)$ having $AD \rightarrow B, C \rightarrow DE, B \rightarrow AE$, and $AE \rightarrow C$, into $R_1(A, B, D)$ and $R_2(A, C, D, E)$ is lossless

- A. Both D_1 and D_2
- B. Neither D_1 nor D_2
- C. Only D_1
- D. Only D_2

ugcnetcse-july2016-paper3 databases

Answer key **7.0.14 UGC NET CSE | December 2011 | Part 2 | Question: 23**

Data security threats include

- A. Privacy invasion
- B. Hardware failure
- C. Fraudulent manipulation of data
- D. Encryption and decryption

ugcnetcse-dec2011-paper2 databases

Answer key 

7.0.15 UGC NET CSE | August 2016 | Part 3 | Question: 12

Which one of the following pairs is correctly matched in the context of database design ?

	List-I(Database term)		List-II(Definition)
I.	Specialization	A.	Result of taking the union of two or more disjoint (lower-level) entity sets to produce a higher-level entity set.
II.	Generalization	B.	Express the number of entities to which another entity can be associated via a relationship set.
III.	Aggregation	C.	Result of taking a subset of a higher-level entity set to form a lower-level entity set.
IV.	Mapping cardinality	D.	An abstraction in which relationship sets (along with their associated entity sets) are treated as higher-level entity sets, and can participate in relationships.

Codes :

A. I-D, II-A, III-B, IV-C
C. I-C, II-D, III-A, IV-B

B. I-D, II-C, III-B, IV-A
D. I-C, II-A, III-D, IV-B

ugcnetcse-aug2016-paper3

Answer key

7.0.16 UGC NET CSE | June 2009 | Part 2 | Question: 18

- (i) DML includes a query language based on both relation algebra and tuple calculus
- (ii) DML includes a query language based on tuple calculus
- (iii) DML includes a query language based on relational algebra
- (iv) DML includes a query language based on none of the relational algebra and tuple calculus

Which one is correct ?

- A. (i) only
- B. (ii) only
- C. (iii) only
- D. (iv) only

ugcnetcse-june2009-paper2

Answer key

7.0.17 UGC NET CSE | June 2009 | Part 2 | Question: 21

Which construct in SQL is used to test whether a subquery has any tuples in its result ?

- A. UNIQUE
- B. EXISTS
- C. GROUP BY
- D. EXCEPT

ugcnetcse-june2009-paper2

Answer key

7.0.18 UGC NET CSE | June 2009 | Part 2 | Question: 22

ORACLE supports :

- A. inner join and outer join only
- B. outer join and semi join only
- C. inner join, outer join , semi join only
- D. inner join, outer join, semi join and anti join

ugcnetcse-june2009-paper2

Answer key 

7.0.19 UGC NET CSE | July 2018 | Part 2 | Question: 68



Database systems that store each relation in a separate operating system file may use the operating system's authorization scheme, instead of defining a special scheme themselves. In this case, which of the following is false?

- A. The administrator enjoys more control on the grant option
- B. It is difficult among the update, delete and insert authorizations
- C. cannot store more than one relation in a file
- D. Operations on the database are speeded up as the authorization procedure is carried out at the operating system level

ugcnetcse-july2018-paper2 databases

Answer key 

7.0.20 UGC NET CSE | December 2004 | Part 2 | Question: 19



The entity type on which the _____ type depends is called the identifying owner.

- A. Strong entity
- B. Relationship
- C. Weak entity
- D. E - R

ugcnetcse-dec2004-paper2

Answer key 

7.0.21 UGC NET CSE | December 2004 | Part 2 | Question: 18



The completeness constraint has rules :

- A. Supertype, Subtype
- B. Total specialization, Partial specialization
- C. Specialization, Generalization
- D. All of the above

ugcnetcse-dec2004-paper2

Answer key 

7.0.22 UGC NET CSE | December 2004 | Part 2 | Question: 17



Specialization is a _____ process.

- A. Top - down
- B. Bottom - up
- C. Both (A) and (B)
- D. None of the above

ugcnetcse-dec2004-paper2

Answer key 

7.0.23 UGC NET CSE | December 2004 | Part 2 | Question: 16



The E-R model is expressed in terms of :

- i. Entities
- ii. The relationship among entities
- iii. The attributes of the entities

Then

- A. (i) and (iii) B. (i) and (ii) C. (ii) and (iii) D. None of the above

ugcnetcse-dec2004-paper2

Answer key 

7.0.24 UGC NET CSE | December 2006 | Part 2 | Question: 20



Two phase protocol in a database management system is :

- A. a concurrency mechanism that is not deadlock free
B. a recovery protocol used for restoring a database after a crash
C. Any update to the system log done in 2-phases
D. not effective in Database

ugcnetcse-dec2006-paper2

Answer key 

7.0.25 UGC NET CSE | December 2006 | Part 2 | Question: 17



In DBMS, deferred update means :

- A. All the updates are done first but the entries are made in the log file later
B. All the log files entries are made first but the actual updates are done later
C. Every update is done first followed by a writing on the log file
D. Changes in the views are deferred till a query asks for a view

ugcnetcse-dec2006-paper2

Answer key 

7.0.26 UGC NET CSE | December 2007 | Part 2 | Question: 20



Consider the query :

```
SELECT student_name FROM students WHERE  
class_name=(SELECT class_name FROM students WHERE math_marks=100);
```

what will be the output ?

- A. the list of names of students with 100 marks in mathematics
B. the names of all students of all classes in which at least one student has 100 marks in mathematics
C. the names of all students in all classes having 100 marks in mathematics
D. the names and class of all students whose marks in mathematics is 100

ugcnetcse-dec2007-paper2

Answer key 

7.0.27 UGC NET CSE | December 2007 | Part 2 | Question: 19



The end of an SQL command is denoted by :

- A. an end-of-line character B. an 'enter-key' marker
C. entering F4 key D. a semicolon (;)

ugcnetcse-dec2007-paper2

Answer key 

7.0.28 UGC NET CSE | December 2007 | Part 2 | Question: 17



Aggregate functions in SQL are :

- A. GREATEST, LEAST and ABS
- B. SUM, COUNT and AVG
- C. UPPER, LOWER and LENGTH
- D. SQRT, POWER and MOD

ugcnetcse-dec2007-paper2

Answer key

7.0.29 UGC NET CSE | December 2007 | Part 2 | Question: 16



A primary key for an entity is :

- A. a candidate key
- B. any attribute
- C. a unique attribute
- D. a superkey

ugcnetcse-dec2007-paper2

Answer key

7.1 Aggregation (2)

7.1.1 Aggregation: UGC NET CSE | December 2008 | Part 2 | Question: 31



Aggregation is :

- A. an abstraction through which relationships are treated as lower level entities
- B. an abstraction through which relationships are treated as higher level entities
- C. an abstraction through which relationships are not treated at all as entities
- D. none of the above

ugcnetcse-dec2008-paper2 databases aggregation

Answer key

7.1.2 Aggregation: UGC NET CSE | July 2016 | Part 3 | Question: 39



Which of the following statements is correct?

- A. Aggregation is a strong type of association between two classes with full ownership
- B. Aggregation is a strong type of association between two classes with partial ownership
- C. Aggregation is a weak type of association between two classes with partial ownership
- D. Aggregation is a weak type of association between two classes with full ownership

ugcnetcse-july2016-paper3 databases aggregation

Answer key

7.2 B Tree (2)

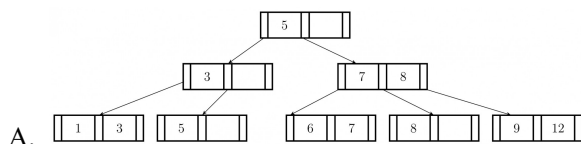
7.2.1 B Tree: UGC NET CSE | January 2017 | Part 3 | Question: 12

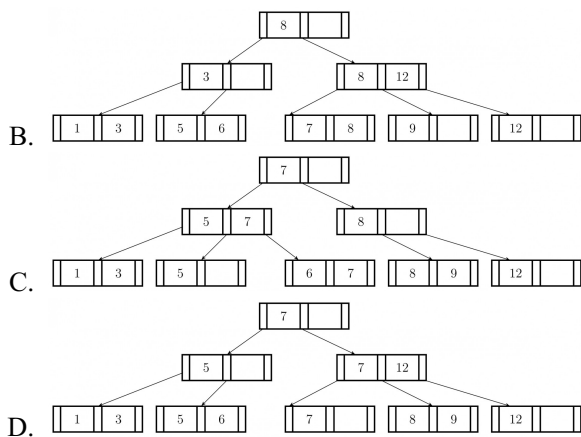


If following sequence of keys are inserted in a B+ tree with $K(=3)$ pointers:

8, 5, 1, 7, 3, 12, 9, 6

Which of the following shall be correct B+ tree?





ugcnetcse-jan2017-paper3 databases b-tree

Answer key

7.2.2 B Tree: UGC NET CSE | June 2014 | Part 2 | Question: 22



The upper bound and lower bound for the number of leaves in a B-tree of degree K with height h is given by :

- A. K^h and $2^{\lceil \frac{K}{2} \rceil h-1}$ B. $K * h$ and $2^{\lceil \frac{K}{2} \rceil h-1}$
 C. K^h and $2^{\lfloor \frac{K}{2} \rfloor h-1}$ D. $K * h$ and $2^{\lfloor \frac{K}{2} \rfloor h-1}$

ugcnetcse-june2014-paper2 databases b-tree

Answer key

7.3 B+tree (1)

7.3.1 B+tree: UGC NET CSE | January 2017 | Part 2 | Question: 20



The order of a leaf node in a B^+ tree is the maximum number of children it can have. Suppose that block size is 1 kilobytes, the child pointer takes 7 bytes long and search field value takes 14 bytes long. The order of the leaf node is _____.

- A. 16 B. 63 C. 64 D. 65

ugcnetjan2017ii databases b+tree

Answer key

7.4 Bcnf Decomposition (1)

7.4.1 Bcnf Decomposition: UGC NET june/August 2024



Which of the following relations can not be decomposed in to BCNF with a lossless join and dependency-preserving decomposition ?

- (1) $R(V, W, X, Y, Z) \{VW \rightarrow X, WX \rightarrow V, VX \rightarrow W, W \rightarrow Y, Y \rightarrow Z\}$
 (2) $R(V, W, X, Y) \{VW \rightarrow W, X \rightarrow Y\}$
 (3) $R(V, W, X, Y) \{VW \rightarrow X, X \rightarrow VY\}$
 (4) $R(V, W, X, Y, Z) \{VW \rightarrow X, X \rightarrow Y, Y \rightarrow Z, Z \rightarrow V\}$

bcnf-decomposition

7.5 Candidate Key (4)

7.5.1 Candidate Key: UGC NET CSE | January 2017 | Part 3 | Question: 8



Which one is correct w.r.t. RDBMS?

- A. primary key $\subseteq$ super key $\subseteq$ candidate key
 B. primary key $\subseteq$ candidate key $\subseteq$ super key

- C. super key $\subseteq$ candidate key $\subseteq$ primary key
 D. super key $\subseteq$ primary key $\subseteq$ candidate key

ugcnetcse-jan2017-paper3 database-normalization candidate-key

Answer key 

7.5.2 Candidate Key: UGC NET CSE | July 2016 | Part 2 | Question: 20



Consider the following database table having A, B, C and D as its four attributes and four possible candidate keys (I, II, III and IV) for this table :

A	B	C	D
a_1	b_1	c_1	d_1
a_2	b_3	c_3	d_1
a_1	b_2	c_1	d_2

I : {B} II : {B, C} III : {A, D} IV : {C, D}

If different symbols stand for different values in the table (e.g., d_1 is definitely not equal to d_2), then which of the above could not be the candidate key for the database table?

- A. I and III only B. III and IV only C. II only D. I only

ugcnetcse-july2016-paper2 databases candidate-key

Answer key 

7.5.3 Candidate Key: UGC NET CSE | June 2019 | Part 2 | Question: 35



Which of the following key constraints is required for functioning of foreign key in the context relational databases?

- A. Unique key B. Primary key C. Candidate key D. Check key

ugcnetcse-june2019-paper2 candidate-key

Answer key 

7.5.4 Candidate Key: UGC NET CSE | June 2019 | Part 2 | Question: 36



In relational database management, which of the following is/are property/properties of candidate key?

P: Uniqueness

Q: Irreducibility

- A. P only B. Q only C. Both P and Q D. Neither P nor Q

ugcnetcse-june2019-paper2 candidate-key

Answer key 

7.6 Cardinality Ratio (1)

7.6.1 Cardinality Ratio: UGC NET CSE | December 2013 | Part 2 | Question: 14



Based on the cardinality ratio and participation _____ associated with a relationship type, choose either the Foreign Key Design, the Cross Referencing Design or Mutual Referencing Design.

- A. Entity B. Constraints C. Rules D. Keys

ugcnetcse-dec2013-paper2 databases cardinality-ratio

Answer key 

7.7 Circular Queue (1)

7.7.1 Circular Queue: UGC NET CSE | December 2008 | Part 2 | Question: 34



If $(rear == maxsize - 1) rear = 0$; else $rear = rear + 1$; is required in :

- A. circular queue
- B. linear queue
- C. stack
- D. deque

ugcnetcse-dec2008-paper2 data-structures circular-queue

Answer key

7.8 Concurrency Control Protocols (1)

7.8.1 Concurrency Control Protocols: UGC NET CSE | December 2009 | Part 2 | Question: 20



Which of the following statement is wrong ?

- I. 2-phase locking protocol suffer from dead lock.
- II. Time stamp protocol suffer from more aborts.
- III. A block hole in a DFD is a data store with only inbound flows.
- IV. Multivalued dependency among attribute is checked at 3 NF level.
- V. An entity-relationship diagram is a tool to represent event model.

- (A) I, II, III
- (B) II, III, IV
- (C) III, IV, V
- (D) II, IV, V

ugcnetcse-dec2009-paper2 concurrency-control-protocols

Answer key

7.9 Conflict Serializable (1)

7.9.1 Conflict Serializable: UGC NET CSE | July 2018 | Part 2 | Question: 65



Consider the following schedules involving two transactions.

$S_1 : r_1(X); r_1(Y); r_2(X); r_2(Y); w_2(Y); w_1(X)$

$S_2 : r_1(X); r_2(X); r_2(Y); w_2(Y); r_1(Y); w_1(X)$

Which one of the following statements is correct with respect to above?

- A. Both S_1 and S_2 are conflict serializable
- B. Both S_1 and S_2 are not conflict serializable
- C. S_1 is conflict serializable and S_2 is not conflict serializable
- D. S_1 is not conflict serializable and S_2 is conflict serializable

ugcnetcse-july2018-paper2 databases transaction-and-concurrency conflict-serializable easy

Answer key

7.10 Crosstabquery (1)

7.10.1 Crosstabquery: UGC NET CSE | September 2013 | Part 2 | Question: 41



Cross_tab displays permit users to view ____ of multidimensional data at a time.

- A. One dimension
- B. Two dimensions
- C. Three dimensions
- D. Multidimensions

Answer key 

7.11

Data Dependency (2)

7.11.1 Data Dependency: UGC NET CSE | December 2006 | Part 2 | Question: 18



Which statement is false regarding data independence ?

- A. Hierarchical data model suffers from data independence
- B. Network model suffers from data independence
- C. Relational model suffers only from logical data independence
- D. Relational model suffers only from physical data independence

ugcnetcse-dec2006-paper2 databases data-dependency

Answer key 

7.11.2 Data Dependency: UGC NET CSE | October 2020 | Part 2 | Question: 14



Consider a relational schema $S = (U, V, W, X, Y, Z)$ on which the following functional dependencies hold:

$(U \rightarrow V, VW \rightarrow X, Y \rightarrow W, X \rightarrow U)$

Which are the candidate keys among following options?

- A. UY, VY
- B. UY, VY, XY
- C. UYZ, VYZ, VWZ
- D. UYZ, VYZ, XYZ

ugcnetcse-oct2020-paper2 databases data-dependency

Answer key 

7.12

Data Integrity (1)

7.12.1 Data Integrity: UGC NET CSE | December 2013 | Part 2 | Question: 15



Data Integrity control uses

- A. Upper and lower limits on numeric data
- B. Passwords to prohibit unauthorized access to files
- C. Data dictionary to keep the data
- D. Data dictionary to find last access data

ugcnetcse-dec2013-paper2 databases data-integrity

Answer key 

7.13

Data Mining (2)

7.13.1 Data Mining: UGC NET CSE | December 2009 | Part 2 | Question: 47



Analysis of large database to retrieve information is called

- (A) OLTP
- (B) OLAP
- (C) OLDP
- (D) OLPP

Answer key **7.13.2 Data Mining: UGC NET CSE | January 2017 | Part 2 | Question: 47**

_____ refers loosely to the process of semi-automatically analyzing large databases to find useful patterns.

- A. Datamining B. Data warehousing C. DBMS D. Data mirroring

ugcnetjan2017ii databases data-mining

Answer key **7.14****Data Model (2)****7.14.1 Data Model: UGC NET CSE | June 2012 | Part 3 | Question: 8, UGCNET-Dec2012-III: 59**

Which level of abstraction describes what data are stored in the database?

- A. Physical level B. View level
C. Abstraction level D. Logical level

ugcnetcse-june2012-paper3 databases data-model ugcnetcse-dec2012-paper3

Answer key **7.14.2 Data Model: UGC NET CSE | September 2013 | Part 2 | Question: 39**

Views are useful for _____ unwanted information, and for collecting together information from more than one relation into a single view.

- A. Hiding B. Deleting C. Highlighting D. All of the above

ugcnetsep2013iii databases data-model

Answer key **7.15****Database Constraints (1)****7.15.1 Database Constraints: UGC NET CSE | December 2013 | Part 2 | Question: 11**

The student marks should not be greater than 100. This is

- A. Integrity constraint B. Referential constraint
C. Over-defined constraint D. Feasible constraint

databases database-constraints ugcnetcse-dec2013-paper2

Answer key **7.16****Database Design (2)****7.16.1 Database Design: UGC NET CSE | December 2005 | Part 2 | Question: 18**

In a relational schema, each tuple is divided in fields called :

- A. Relations B. Domains C. Queries D. All the above

ugcnetcse-dec2005-paper2 databases database-design

7.16.2 Database Design: UGC NET CSE | January 2017 | Part 2 | Question: 16

An attribute A of datatype varchar (20) has value 'Ram' and the attribute B of datatype char (20) has value 'Sita' in oracle. The attribute A has _____ memory spaces and B has _____ memory spaces.

- A. 20,20 B. 3,20 C. 3,4 D. 20,4

ugcnetjan2017ii databases database-design

7.17.1 Database Normalization: UGC NET CSE | December 2004 | Part 2 | Question: 20



Match the following :

- | | |
|-----------|---|
| (i) 2NF | (a) Transitive dependencies eliminated |
| (ii) 3NF | (b) Multivalued attribute removed |
| (iii) 3NF | (c) Contains no partial functional dependencies |
| (iv) 5NF | (d) Contains no join dependency |

A. i-a, ii-c, iii-b, iv-d
C. i-d, ii-c, iii-b, iv-a

B. i-d, ii-c, iii-a, iv-b
D. i-a, ii-b, iii-c, iv-d

ugcnetcse-dec2004-paper2 databases database-normalization match-the-following easy

7.17.2 Database Normalization: UGC NET CSE | December 2008 | Part 2 | Question: 32



Suppose R is a relation schema and F is a set of functional dependencies on R. Further, suppose R1 and R2 form a decomposition of R. Then the decomposition is a lossless join decomposition of R provided that:

- A. $R1 \cap R2 \rightarrow R1$ is in F^+
B. $R1 \cap R2 \rightarrow R2$ is in F^+
C. both $R1 \cap R2 \rightarrow R1$ and $R1 \cap R2 \rightarrow R2$ functional dependencies are in F^+
D. at least one from $R1 \cap R2 \rightarrow R1$ and $R1 \cap R2 \rightarrow R2$ is in F^+

ugcnetcse-dec2008-paper2 databases database-normalization

7.17.3 Database Normalization: UGC NET CSE | December 2009 | Part 2 | Question: 18



Match the following :

- | | |
|---------------------------|--|
| (1) Determinants | (a) No attribute can be added |
| (2) Candidate key | (b) Uniquely identified a row |
| (3) Non-redundancy | (c) A constraint between two attribute |
| (4) Functional Dependency | (d) Group of attributes on the left hand side of arrow of function dependency. |

A. 1-d, 2-b, 3-a, 4-c
C. 4-a, 3-b, 2-c, 1-d

B. 2-d, 3-a, 1-b, 4-c
D. 3-a, 4-b, 1-c, 2-d

ugcnetcse-dec2009-paper2 databases database-normalization match-the-following

7.17.4 Database Normalization: UGC NET CSE | December 2009 | Part 2 | Question: 19



A function that has no partial functional dependencies is in _____ form.

- (A) 3 NF
(B) 2 NF
(C) 4 NF
(D) BCNF

ugcnetcse-dec2009-paper2 databases database-normalization

7.17.5 Database Normalization: UGC NET CSE | December 2010 | Part 2 | Question: 17

The dependency preservation decomposition is a property to decompose database schema D , in which each functional dependency $X \rightarrow Y$ specified in F ,

- A. Appeared directly in one of the relation schemas R_i in the decomposed D .
- B. Could be inferred from dependencies that appear in some R_i .
- C. Both (A) and (B)
- D. None of these

ugcnetcse-dec2010-paper2 databases database-normalization

Answer key

7.17.6 Database Normalization: UGC NET CSE | December 2012 | Part 2 | Question: 13

Which of the following is true?

- A. A relation in BCNF is always in 3NF
- B. A relation in 3NF is always in BCNF
- C. BCNF and 3NF are same
- D. A relation in BCNF is not in 3NF

databases ugcnetcse-dec2012-paper2 database-normalization

Answer key

7.17.7 Database Normalization: UGC NET CSE | December 2012 | Part 3 | Question: 63

The third normal form is based on the concept of _____

- A. Closure Dependency
- B. Transitive Dependency
- C. Normal Dependency
- D. Functional Dependency

ugcnetcse-dec2012-paper3 databases database-normalization

Answer key

7.17.8 Database Normalization: UGC NET CSE | December 2013 | Part 3 | Question: 59

Armstrong (1974) proposed systematic approach to derive functional dependencies. Match the following w.r.t functional dependencies:

List-I**List -II**

- | | |
|-----------------------------|---|
| a. Decomposition Rule | i. If $X \rightarrow Y$ and $Z \rightarrow W$ then $\{X, Z\} \rightarrow \{Y, W\}$ |
| b. Union rule | ii. If $X \rightarrow Y$ and $\{Y, W\} \rightarrow Z$ then $\{X, W\} \rightarrow Z$ |
| c. Composition rule | iii. If $X \rightarrow Y$ and $X \rightarrow Z$ then $X \rightarrow \{Y, Z\}$ |
| d. Pseudo transitivity rule | iv. If $X \rightarrow \{Y, Z\}$ then $X \rightarrow Y$ and $X \rightarrow Z$ |

Codes:

- A. a-iii, b-ii, c-iv, d-i
- B. a-i, b-iii, c-iv, d-ii
- C. a-ii, b-i, c-iii, d-iv
- D. a-iv, b-iii, c-i, d-ii

ugcnetcse-dec2013-paper3 databases database-normalization

Answer key

7.17.9 Database Normalization: UGC NET CSE | December 2014 | Part 2 | Question: 19

The best normal form of relation scheme $R(A, B, C, D)$ along with the set of functional dependencies $F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$ is

- A. Boyce-Codd Normal form
- B. Third Normal form
- C. Second Normal form
- D. First Normal form

ugcnetcse-dec2014-paper2 databases database-normalization

Answer key

7.17.10 Database Normalization: UGC NET CSE | December 2014 | Part 2 | Question: 20

Identify the minimal key for relational scheme $R(A, B, C, D, E)$ with functional dependencies $F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$

- A. A B. AE C. BE D. CE

ugcnetcse-dec2014-paper2 databases database-normalization

Answer key

7.17.11 Database Normalization: UGC NET CSE | December 2014 | Part 3 | Question: 10

Let $R = ABCDE$ is a relational scheme with functional dependency set $F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$. The attribute closures of A and E are

- A. $ABCD, \phi$ B. $ABCD, E$
C. Φ, ϕ D. ABC, E

ugcnetcse-dec2014-paper3 databases database-normalization

Answer key

7.17.12 Database Normalization: UGC NET CSE | December 2014 | Part 3 | Question: 12

Which of the following is false ?

- A. Every binary relation is never be in $BCNF$.
B. Every $BCNF$ relation is in $3NF$.
C. $1NF, 2NF, 3NF$ and $BCNF$ are based on functional dependencies.
D. Multivalued Dependency (MVD) is a special case of Join Dependency (JD)

ugcnetcse-dec2014-paper3 databases database-normalization

Answer key

7.17.13 Database Normalization: UGC NET CSE | December 2015 | Part 2 | Question: 20

A relation $R = \{A, B, C, D, E, F, G\}$ is given with following set of functional dependencies: $F = \{AD \rightarrow E, BE \rightarrow F, B \rightarrow C, AF \rightarrow G\}$. Which of the following is a candidate key?

- A. A B. AB C. ABC D. ABD

ugcnetcse-dec2015-paper2 databases database-normalization

Answer key

7.17.14 Database Normalization: UGC NET CSE | January 2017 | Part 3 | Question: 10

For database relation $R(A, B, C, D)$ where the domains of A, B, C and D include only atomic values, only the following functional dependencies and those that can be inferred from them are:

$A \rightarrow C$
 $B \rightarrow D$

The relation R is in

- A. First normal form but not in second normal form
B. Both in first normal form as well as in second normal form
C. Second normal form but not in third normal form.
D. Both in second normal form as well as in third normal form.

ugcnetcse-jan2017-paper3 databases database-normalization

Answer key

7.17.15 Database Normalization: UGC NET CSE | June 2005 | Part 2 | Question: 19

Multi-valued dependency among attribute is checked at which level ?

- A. 2NF B. 3NF C. 4NF D. 5NF

ugcnetcse-june2005-paper2 database-normalization

Answer key

7.17.16 Database Normalization: UGC NET CSE | June 2010 | Part 2 | Question: 18

Match the following :

- | | |
|--------|---|
| a. 2NF | i. Transitive dependencies eliminated |
| b. 3NF | ii. Multivalued attribute removed |
| c. 4NF | iii. Contain no partial Functional dependencies |
| d. 5NF | iv. Contains no join dependencies |

Codes :

- | | |
|---------------------------|---------------------------|
| A. a-i, b-iii, c-ii, d-iv | B. a-ii, b-iii, c-iv, d-i |
| C. a-iii, b-iv, c-i, d-ii | D. a-iii, b-ii, c-iv, d-i |

ugcnetcse-june2010-paper2 databases database-normalization

Answer key

7.17.17 Database Normalization: UGC NET CSE | June 2013 | Part 3 | Question: 9

For a database relation $R(a, b, c, d)$ where the domains of a, b, c, d include only the atomic values. The functional dependency $a \rightarrow c, b \rightarrow d$ holds in the following relation.

- A. In 1NF not in 2NF B. In 2NF not in 3NF C. In 3NF D. in 1 NF

ugcnetcse-june2013-paper3 databases database-normalization

Answer key

7.17.18 Database Normalization: UGC NET CSE | June 2014 | Part 2 | Question: 49

Let $R = \{A, B, C, D, E, F\}$ be a relation schema with the following dependencies $C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B$. Which of the following is a key for R ?

- A. CD B. EC C. AE D. AC

ugcnetcse-june2014-paper2 databases database-normalization

Answer key

7.17.19 Database Normalization: UGC NET CSE | June 2014 | Part 3 | Question: 22

Consider the following relational schemas for a library database :

Book (Title, Author, Catalog_no, Publisher, Year, Price) Collection(Title, Author, Catalog_no) with the following functional dependencies :

I. Title, Author $\rightarrow$ Catalog_no

II. Catalog_no $\rightarrow$ Title, Author, Publisher, Year

III. Publisher, Title, Year $\rightarrow$ Price Assume (Author, Title) is the key for both schemas. Which one of the following is true ?

- | | |
|--|---|
| A. Both Book and Collection are in BCNF. | B. Both Book and Collection are in 3NF. |
| C. Book is in 2NF and Collection in 3NF. | D. Both Book and Collection are in 2NF. |

ugcnetjune2014iii databases database-normalization

Answer key

7.17.20 Database Normalization: UGC NET CSE | June 2019 | Part 2 | Question: 34

In relational databases, if relation R is in BCNF, then which of the following is true about relation R ?

- A. R is in 4NF
 B. R is not in 1NF
 C. R is in 2NF and not in 3NF
 D. R is in 2NF and 3NF

ugcnetcse-june2019-paper2 database-normalization

Answer key

7.17.21 Database Normalization: UGC NET CSE | Junet 2015 | Part 3 | Question: 10

The Relation

Vendor Order (V_no , V_ord_no , V_name , Qty_sup , $unit_price$) is in 2NF because

- A. Non_key attribute V_name is dependent on V_no which is part of composite key
 B. Non_key attribute V_name is dependent on Qty_sup
 C. Key attribute Qty_sup is dependent on primary key $unit_price$
 D. Key attribute V_ord_no is dependent on primary key $unit_price$

ugcnetcse-june2015-paper3 databases database-normalization

Answer key

7.17.22 Database Normalization: UGC NET CSE | November 2017 | Part 3 | Question: 9

If every non-key attribute is functionally dependent on the primary key, then the relation is in _____

- A. First normal form
 B. Second normal form
 C. Third normal form
 D. Fourth normal form

ugcnetcse-nov2017-paper3 database-normalization

Answer key

7.17.23 Database Normalization: UGC NET CSE | October 2022 | Part 1 | Question: 53

Match **List I** with **List II** :

List I List II

- (A) BCNF (I) It removes multivalued dependency
 (B) 3NF (II) It is not always dependency preserving
 (C) 2NF (III) It removes transitive dependency
 (D) 4NF (IV) It removes partial functional dependency

Choose the correct answer from the options given below:

- A. (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
 B. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
 C. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
 D. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

ugcnetcse-oct2022-paper1 database-normalization

Answer key

7.18**Decomposition (3)****7.18.1 Decomposition: UGC NET CSE | August 2016 | Part 3 | Question: 10**

Consider the table R with attributes A, B and C . The functional dependencies that hold on R are : $A \rightarrow B, C \rightarrow AB$. Which of the following statements is/are True ?

- I. The decomposition of R into $R_1(C, A)$ and $R_2(A, B)$ is lossless.
 II. The decomposition of R into $R_1(A, B)$ and $R_2(B, C)$ is lossy.

- A. Only I B. Only II C. Both I and II D. Neither I nor II

Answer key **7.18.2 Decomposition: UGC NET CSE | December 2012 | Part 3 | Question: 56**

If a relation with a Schema R is decomposed into two relations R_1 and R_2 such that $(R_1 \cup R_2) = R$ then which one of the following is to be satisfied for a lossless joint decomposition ($\rightarrow$ indicates functional independency)

- A. $(R_1 \cap R_2) \rightarrow R_1$ or $R_1 \cap R_2 \rightarrow R_2$
- B. $R_1 \cap R_2 \rightarrow R_1$
- C. $R_1 \cap R_2 \rightarrow R_2$
- D. $(R_1 \cap R_2) \rightarrow R_1$ and $R_1 \cap R_2 \rightarrow R_2$

ugcnetcse-dec2012-paper3 databases decomposition

Answer key **7.18.3 Decomposition: UGC NET CSE | Junet 2015 | Part 3 | Question: 11**

The relation schemas R_1 and R_2 form a Lossless join decomposition of R if and only if

- i. $R_1 \cap R_2 \rightarrow (R_1 - R_2)$
 - ii. $R_1 \rightarrow R_2$
 - iii. $R_1 \cap R_2 \rightarrow (R_2 - R_1)$
 - iv. $R_2 \rightarrow R_1 \cap R_2$
- A. i and ii happen B. i and iv happen
 C. i and iii happen D. ii and iii happen

ugcnetcse-june2015-paper3 databases decomposition database-normalization

Answer key **7.19 Dependency Preserving (1)****7.19.1 Dependency Preserving: UGC NET CSE | January 2017 | Part 2 | Question: 19**

Consider a schema $R(MNPQ)$ and functional dependencies $M \rightarrow N, P \rightarrow Q$. Then the decomposition of R into $R_1(MN)$ and $R_2(PQ)$ is _____.

- A. Dependency preserving but not lossless join
- B. Dependency preserving and lossless join
- C. Lossless join but not dependency preserving
- D. Neither dependency preserving nor lossless join.

ugcnetjan2017ii databases dependency-preserving

Answer key **7.20 Distributed Database (2)****7.20.1 Distributed Database: UGC NET CSE | August 2016 | Part 3 | Question: 8**

The Global conceptual Schema in a distributed database contains information about global relations. The condition that all the data of the global relation must be mapped into the fragments, that is, it must not happen that a data item which belongs to a global relation does not belong to any fragment, is called :

- A. Disjointness condition
- B. Completeness condition
- C. Reconstruction condition
- D. Aggregation condition

ugcnetcse-aug2016-paper3 databases distributed-database

Answer key 

7.20.2 Distributed Database: UGC NET CSE | July 2016 | Part 3 | Question: 8



In distributed databases, location transparency allows for database users, programmers and administrators to treat the data as if it is at one location, A SQL query with location transparency needs to specify

- A. Inheritances B. Fragments C. Locations D. Local formats

ugcnetcse-july2016-paper3 databases distributed-database

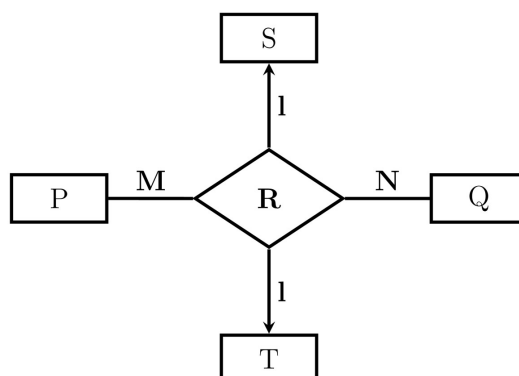
Answer key

7.21 ER Diagram (18)

7.21.1 ER Diagram: UGC NET CSE | August 2016 | Part 2 | Question: 19



Consider the following Entity-Relationship ($E - R$) diagram and three possible relationship sets I, II and III) for this $E - R$ diagram:



I :

P	Q	S	T
p_1	q_1	s_1	t_1
p_1	q_1	s_1	t_2

II :

P	Q	S	T
p_1	q_1	s_1	t_1
p_1	q_1	s_2	t_2

III :

P	Q	S	T
p_1	q_1	s_1	t_1
p_1	q_2	s_1	t_1

If different symbols stand for different values (e.g., t_1 is definitely not equal to t_2), then which of the above could not be the relationship set for the $E - R$ diagram ?

- A. I only B. I and II only
C. II only D. I, II and III

ugcnetcse-aug2016-paper2 databases er-diagram

Answer key

7.21.2 ER Diagram: UGC NET CSE | December 2008 | Part 2 | Question: 30



An entity has:

- (i) a set of properties
(ii) a set of properties and values for all the properties
(iii) a set of properties and the values for some set of properties may non-uniquely identify an entity
(iv) a set of properties and the values for some set of properties may uniquely identify an entity

Which of the above are valid?

- A. (i) only
B. (ii) only
C. (iii) only
D. (iv) only

ugcnetcse-dec2008-paper2 databases er-diagram

Answer key

7.21.3 ER Diagram: UGC NET CSE | December 2009 | Part 2 | Question: 16



The E-R model is expressed in term of

- I. Entities
- II. The relationship among entities.
- III. The attributes of the entities.
- IV. Functional relationship.

- (A) I, II
- (B) I, II, IV
- (C) II, II, IV
- (D) I, II, III

ugcnetcse-dec2009-paper2 databases er-diagram

Answer key

7.21.4 ER Diagram: UGC NET CSE | December 2009 | Part 2 | Question: 17



Specialization is _____ process.

- (A) top-down
- (B) bottom up
- (C) both (A) and (B)
- (D) none of these

ugcnetcse-dec2009-paper2 databases er-diagram

Answer key

7.21.5 ER Diagram: UGC NET CSE | December 2010 | Part 2 | Question: 16



In generalisation, the differences between members of an entity is

- A. Maximized
- B. Minimized
- C. Both (A) & (B)
- D. None of these

ugcnetcse-dec2010-paper2 databases er-diagram

Answer key

7.21.6 ER Diagram: UGC NET CSE | December 2013 | Part 2 | Question: 13



An E-R model includes

- I. An ER diagram portraying entity types.
- II. Attributes for each entity type.
- III. Relationships among entity types.
- IV. Semantic integrity constraints that reflects the business rules about data not captured in the ER diagram

- A. I, II, III and IV
- B. I and IV
- C. I, II and IV
- D. I and III

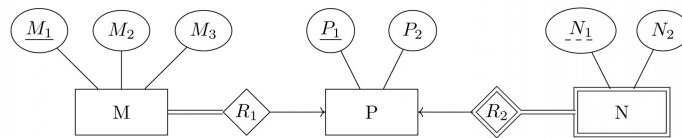
ugcnetcse-dec2013-paper2 databases er-diagram

Answer key

7.21.7 ER Diagram: UGC NET CSE | December 2013 | Part 3 | Question: 55



Consider the following ER diagram:



The minimum number of tables required to represent M, N, P, R_1, R_2 is

- A. 2 B. 3 C. 4 D. 5

ugcnetcse-dec2013-paper3 databases er-diagram

Answer key

7.21.8 ER Diagram: UGC NET CSE | December 2013 | Part 3 | Question: 57



Find the false statement:

- A. The relationship construct known as the weak relationship type was defined by Dey, Storey & Barron (1999)
 B. A weak relationship occurs when two relationship types are linked by either Event- Precedent sequence or Condition- recedent sequence.
 C. Conceptual model is not accurate representation of “Universe of interest”.
 D. Ternary, Quaternary and Quintary relationships are shown through a series of application scenario’s and vignette’s.

ugcnetcse-dec2013-paper3 databases er-diagram

Answer key

7.21.9 ER Diagram: UGC NET CSE | December 2014 | Part 2 | Question: 18



What kind of mechanism is to be taken into account for converting a weak entity set into strong entity set in entity-relationship diagram

- A. Generalization B. Aggregation
 C. Specialization D. Adding suitable attributes

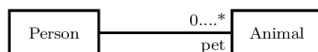
ugcnetcse-dec2014-paper2 databases er-diagram

Answer key

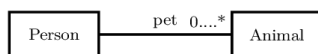
7.21.10 ER Diagram: UGC NET CSE | December 2014 | Part 3 | Question: 38



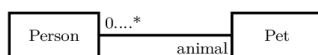
How to express that some person keeps animals as pets?



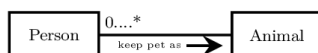
A.



B.



C.



D.

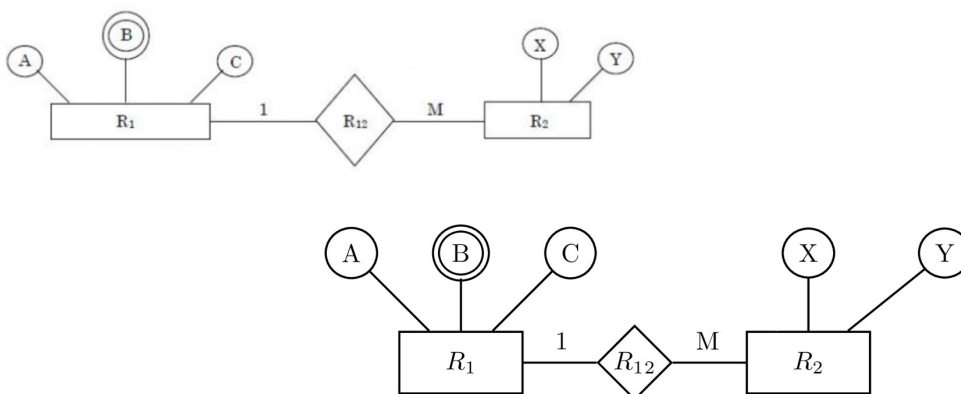
ugcnetcse-dec2014-paper3 databases er-diagram

Answer key

7.21.11 ER Diagram: UGC NET CSE | December 2019 | Part 2 | Question: 23



Find minimum number of tables required for converting the following entity relationship diagram into relational database?



- A. 2 B. 4 C. 3 D. 5

ugcnetcse-dec2019-paper2 er-diagram

Answer key

7.21.12 ER Diagram: UGC NET CSE | January 2017 | Part 2 | Question: 18



Let M and N be two entities in an $E - R$ diagram with simple single value attributes. R_1 and R_2 are two relationship between M and N , where as R_1 , is one-to-many and R_2 , is many-to-many.

The minimum number of tables required to represent M, N, R_1 , and R_2 , in the relational model are _____.

- A. 4 B. 6 C. 7 D. 3

ugcnetjan2017ii databases er-diagram

Answer key

7.21.13 ER Diagram: UGC NET CSE | July 2018 | Part 2 | Question: 64



Relations produced from E-R Model will always be in _____

- A. 1 NF B. 2 NF C. 3 NF D. 4 NF

databases er-diagram database-normalization ugcnetcse-july2018-paper2

Answer key

7.21.14 ER Diagram: UGC NET CSE | June 2010 | Part 2 | Question: 16



An entity instance is a single occurrence of an _____.

- A. Entity type B. Relationship type
C. Entity and relationship type D. None of these

ugcnetcse-june2010-paper2 databases er-diagram

Answer key

7.21.15 ER Diagram: UGC NET CSE | Junet 2015 | Part 2 | Question: 49



Which of the following statements is false about weak entity set?

- A. Weak entities can be deleted automatically when their strong entity is deleted
B. Weak entity set avoids the data duplication and consequent possible inconsistencies caused by duplicating the key of the string entity
C. A weak entity set has no primary keys unless attributes of the strong entity set on which it depends are included

D. Tuples in a weak entity set are not partitioned according to their relationship with tuples with a strong entity set

ugcnetcse-june2015-paper2 er-diagram databases

Answer key 

7.21.16 ER Diagram: UGC NET CSE | Junet 2015 | Part 3 | Question: 7



Let E_1 and E_2 be two entities in E-R diagram with simple single valued attributes. R_1 and R_2 are two relationships between E_1 and E_2 where R_1 is one-many and R_2 is many-many. R_1 and R_2 do not have any attribute of their own. How many minimum number of tables are required to represent this situation in the Relational Model?

- A. 4 B. 3 C. 2 D. 1

databases er-diagram ugcnetcse-june2015-paper3

Answer key 

7.21.17 ER Diagram: UGC NET CSE | September 2013 | Part 3 | Question: 48



Match the following :

- | | |
|---------------|-------------------------|
| (a) Create | (i) The ER model |
| (b) Select | (ii) Relationship model |
| (c) Rectangle | (iii) DDL |
| (d) Record | (iv) DML |

Codes :

- | | |
|---|---|
| A. (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii) | B. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i) |
| C. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii) | D. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i) |

ugcnetcse-sep2013-paper3 databases er-diagram match-the-following easy

Answer key 

7.21.18 ER Diagram: UGC NET CSE | September 2013 | Part 3 | Question: 49



Match the following:

- | | |
|--|--------------------------------|
| a.  | i. One to One Relationship |
| b.  | ii. Relationship |
| c.  | iii. Many to Many Relationship |
| d.  | iv. Many to One Relationship |

- | | |
|---------------------------|---------------------------|
| A. a-iii, b-iv, c-ii, d-i | B. a-iv, b-iii, c-ii, d-i |
| C. a-ii, b-iii, c-iv, d-i | D. a-iii, b-iv, c-i, d-ii |

ugcnetcse-sep2013-paper3 databases er-diagram match-the-following easy

Answer key 

7.22 Enhanced ER Model (1)

7.22.1 Enhanced ER Model: UGC NET CSE | June 2014 | Part 3 | Question: 23



Specialization Lattice stands for

- A. An entity type can participate as a subclass in only one specialization.
- B. An entity type can participate as a subclass in more than one specialization.
- C. An entity type that can participate in one specialization.
- D. An entity type that can participate in one generalization.

ugcnetjune2014iii databases enhanced-er-model

Answer key 

7.23

File System (1)

7.23.1 File System: UGC NET CSE | Junet 2015 | Part 2 | Question: 19



Database application were directly built on top of file system to overcome the following drawbacks of using file-systems:

- i. Data redundancy and inconsistency
- ii. Difficulty in accessing data
- iii. Data isolation
- iv. Integrity problems

- A. i
- B. i and iv
- C. i, ii, and iii
- D. i, ii, iii, and iv

ugcnetcse-june2015-paper2 databases file-system

Answer key 

7.24

Functional Dependency (1)

7.24.1 Functional Dependency: UGC NET CSE | January 2017 | Part 3 | Question: 9



Let $pk(R)$ denotes primary key of relation R . A many-to-one relationship that exists between two relations R_1 and R_2 can be expressed as follows:

- A. $pk(R_2) \rightarrow pk(R_1)$
- B. $pk(R_1) \rightarrow pk(R_2)$
- C. $pk(R_2) \rightarrow R_1 \cap R_2$
- D. $pk(R_1) \rightarrow R_1 \cap R_2$

ugcnetcse-jan2017-paper3 databases functional-dependency database-normalization

Answer key 

7.25

Generalization (1)

7.25.1 Generalization: UGC NET CSE | June 2010 | Part 2 | Question: 17



Generalization is _____ process.

- A. Top-down
- B. Bottom up
- C. Both (A) & (B)
- D. None of these

ugcnetcse-june2010-paper2 databases generalization

Answer key 

7.26

Granularity (1)

7.26.1 Granularity: UGC NET CSE | June 2012 | Part 3 | Question: 4



What is Granularity?

- A. The size of database
- B. The size of data item
- C. The size of record
- D. The size of file

ugcnetcse-june2012-paper3 databases granularity

Answer key 

7.27

Hierarchical Database (1)

7.27.1 Hierarchical Database: UGC NET CSE | July 2018 | Part 2 | Question: 63



In a Hierarchical database, a hashing function is used to locate the ____

- A. Collision B. Root C. Foreign Key D. Records

ugcnetcse-july2018-paper2 databases hierarchical-database

Answer key

7.28 Indexing (1)

7.28.1 Indexing: UGC NET CSE | June 2014 | Part 2 | Question: 48



A clustering index is created when ____.

- A. Primary key is declared and ordered B. No key ordered
C. Foreign key ordered D. There is no key and no order

ugcnetcse-june2014-paper2 databases indexing

Answer key

7.29 Java (1)

7.29.1 Java: UGC NET CSE | December 2012 | Part 2 | Question: 22



In DML, RECONNECT command cannot be used with

- A. OPTIONAL Set B. FIXED Set C. MANDATOR Set D. All of the above

ugcnetcse-dec2012-paper2 programming java bad-question

Answer key

7.30 Joins (2)

7.30.1 Joins: UGC NET CSE | December 2015 | Part 3 | Question: 63



Consider the following three tables R , S and T . In this question, all the join operations are natural joins ($\bowtie$). (π) is the projection operation of a relation:

R		S		T	
A	B	B	C	A	C
1	2	6	2	7	1
3	2	2	4	1	2
5	6	8	1	9	3
7	8	8	3	5	4
9	8	2	5	3	5

Possible answer tables for this question are also given as below:

A	B	C
1	2	4
1	2	5
3	2	4
3	2	5
5	6	2
7	8	1
7	8	3
9	8	1
9	8	3

(a)

A	B	C
1	2	2
3	2	5
5	6	4
7	8	1
9	8	3

(b)

A	B	C
1	6	2
3	2	5
5	2	4
7	8	1
9	8	3

(c)

A	B	C
3	2	5
7	8	1
9	8	3

(d)

A	B	C
1	2	4
1	2	5
3	2	4
3	2	5
5	6	2
7	8	1
7	8	3
9	8	1
9	8	3

(a)

A	B	C
1	2	2
3	2	5
5	6	4
7	8	1
9	8	3

(b)

A	B	C
1	6	2
3	2	5
5	2	4
7	8	1
9	8	3

(c)

A	B	C
3	2	5
7	8	1
9	8	3

(d)

- A. (a) B. (b) C. (c) D. (d)

ugcnetcse-dec2015-paper3 databases joins

Answer key 

7.30.2 Joins: UGC NET CSE | July 2018 | Part 2 | Question: 61



In RDBMS, which type of Join returns all rows that satisfy the join condition?

- A. Inner Join B. Outer Join
C. Semi Join D. Anti Join

ugcnetcse-july2018-paper2 databases rdbms joins

Answer key 

7.31

Object Oriented Database (2)

7.31.1 Object Oriented Database: UGC NET CSE | December 2015 | Part 3 | Question: 58



Which of the following statements regarding the features of the object-oriented approach to databases are true?

- i. The ability to develop more realistic models of the real world
ii. The ability to represent the world in a non-geometric way
iii. The ability to develop databases using natural language approaches
iv. The need to split objects into their component parts
v. The ability to develop database models based on location rather than state and behaviour
- A. i, ii and iii B. ii, iii and iv
C. i, iv, and v D. iii, iv, and v

ugcnetcse-dec2015-paper3 databases object-oriented-database

Answer key 

7.31.2 Object Oriented Database: UGC NET CSE | July 2016 | Part 3 | Question: 7



Which of the following statements concerning Object-oriented database is false?

- A. Objects in object-oriented database contain not only data but also methods for processing the data
B. Object-oriented database store computational instructions in the same place as the data
C. Object-oriented database are mmore adapt at handling structured (analytical) data than relational databases
D. Object-oriented databases store more types of data than relational databases and access that data faster

ugcnetcse-july2016-paper3 databases object-oriented-database

7.32

Protocol (1)

7.32.1 Protocol: UGC NET CSE | June 2014 | Part 3 | Question: 27, UGCNET-June2015-II: 17



Which of the following concurrency protocol ensures both conflict serializability and freedom from deadlock:

- I. 2-phase locking
II. Time phase ordering

- A. Both I & II B. II only C. I only D. Neither I nor II

Answer key 

7.33

Question Fixed (1)

7.33.1 Question Fixed: UGC NET CSE | June 2023 | Part 2: 19



Let $R(A, B, C, D, E, F)$ be a relational schema with following functional dependencies:
 $C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B$. Which of the following is a key for R ?

- A. CD B. EC C. AE D. AC

ugcnetcse-june2023-paper2 database-normalization question-fixed

Answer key 

7.34

Rdbms (8)

7.34.1 Rdbms: UGC NET CSE | July 2016 | Part 2 | Question: 16



In RDBMS, the constraint that no key attribute (column) may be NULL is referred to as:

- A. Referential integrity B. Multi-valued dependency
 C. Entity integrity D. Functional dependency

ugcnetcse-july2016-paper2 databases rdbms sql

Answer key 

7.34.2 Rdbms: UGC NET CSE | July 2016 | Part 2 | Question: 18



In a relational database model, NULL values can be used for all but which one of the following?

- A. To allow duplicate tuples in the table by filling the primary key column(s) with NULL
 B. To avoid confusion with actual legitimate data values like 0 (zero) or integer columns and " (the empty string) for string columns
 C. To leave columns in a tuple marked as "unknown" when the actual value is unknown
 D. To fill a column in a tuple when that column does not really "exist" for that particular tuple

ugcnetcse-july2016-paper2 rdbms sql databases

Answer key 

7.34.3 Rdbms: UGC NET CSE | July 2018 | Part 2 | Question: 62



Consider a relation book (title, price) which contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query?

```
Select title
from book as B
where (select count(*)
      from book as T
      where T.price > B.price < 7
```

- A. Titles of the six most expensive books
 B. Title of the sixth most expensive books
 C. Titles of the seven most expensive books
 D. Title of the seventh most expensive books

ugcnetcse-july2018-paper2 databases rdbms sql

Answer key 

7.34.4 Rdbms: UGC NET CSE | July 2018 | Part 2 | Question: 66

For a database relation $R(a, b, c, d)$ where the domains of a, b, c and d include only atomic values and only the following functional dependencies and those that can be inferred from them hold:

$$a \rightarrow c$$

$$b \rightarrow d$$

The relation is in _____

- A. First normal form but not in second normal form
- B. Second normal form but not in third normal form
- C. Third normal form
- D. BCNF

ugcnetcse-july2018-paper2 databases rdbms

Answer key

7.34.5 Rdbms: UGC NET CSE | July 2018 | Part 2 | Question: 67

A many-to-one relationship exists between entity sets r_1 and r_2 . How will it be represented using functional dependencies if $Pk(r)$ denotes the primary key attribute of relation r ?

- A. $Pk(r_1) \rightarrow Pk(r_2)$
- B. $Pk(r_2) \rightarrow Pk(r_1)$
- C. $Pk(r_2) \rightarrow Pk(r_1)$ and $Pk(r_1) \rightarrow Pk(r_2)$
- D. $Pk(r_2) \rightarrow Pk(r_1)$ or $Pk(r_1) \rightarrow Pk(r_2)$

ugcnetcse-july2018-paper2 databases rdbms

Answer key

7.34.6 Rdbms: UGC NET CSE | June 2013 | Part 3 | Question: 34

Horn clauses are special kinds of propositions which can be described as

- A. Single atomic proposition on left side
- B. Single or multiple atomic proposition on left side
- C. A single atomic proposition on left side and a single atomic proposition on right side
- D. A single atomic proposition on left side or an empty left side

ugcnetcse-june2013-paper3 databases rdbms

Answer key

7.34.7 Rdbms: UGC NET CSE | June 2013 | Part 3 | Question: 7

The "PROJECT" operator of a relational algebra creates a new table that has always

- A. More columns than columns in original table
- B. More rows than original table
- C. Same number of rows as the original table
- D. Same number of columns as the original table

ugcnetcse-june2013-paper3 databases rdbms

Answer key

7.34.8 Rdbms: UGC NET CSE | November 2017 | Part 3 | Question: 10

Consider a relation $R(A, B, C, D, E, F, G, H)$ where each attribute is atomic, and following functional dependencies exist.

$$CH \rightarrow G$$

$$A \rightarrow BC$$

$$B \rightarrow CFH$$

$$E \rightarrow A$$

$$F \rightarrow EG$$

The relation R is _____

- A. In $1NF$ but not in $2NF$
- B. In $2NF$ but not in $3NF$
- C. In $3NF$ but not in $BCNF$
- D. In $BCNF$

ugcnetcse-nov2017-paper3 databases rdbms database-normalization

Answer key 

7.35

Recovery From Failure (1)

7.35.1 Recovery From Failure: UGC NET CSE | December 2014 | Part 3 | Question: 09



_____ rules used to limit the volume of log information that has to be handled and processed in the event of system failure involving the loss of volatile information.

- A. Write-ahead log
- B. Check-pointing
- C. Log buffer
- D. Thomas

ugcnetcse-dec2014-paper3 databases transaction-and-concurrency recovery-from-failure

Answer key 

7.36

Referential Integrity (3)

7.36.1 Referential Integrity: UGC NET CSE | December 2012 | Part 3 | Question: 70



Referential integrity is directly related to

- A. Relation key
- B. Foreign key
- C. Primary key
- D. Candidate key

ugcnetcse-dec2012-paper3 databases relational-model referential-integrity candidate-key

Answer key 

7.36.2 Referential Integrity: UGC NET CSE | June 2019 | Part 2 | Question: 39



Following table has two attributes Employee_id and Manager_id, where Employee_id is a primary key and manager_id is a foreign key referencing Employee_id with on-delete cascade:

Employee_id	Manager_id
20	40
25	40
30	35
35	20
40	45
45	25

On deleting the table (20, 40), the set of other tuples that must be deleted to maintain the referential integrity of table is

- A. (30,35) only
- B. (30,35) and (35,20) only
- C. (35,20) only
- D. (40,45) and (25,40) only

ugcnetcse-june2019-paper2 referential-integrity

Answer key 

7.36.3 Referential Integrity: UGC NET CSE | September 2013 | Part 3 | Question: 24



_____ constraints ensure that a value that appears in one relation for a given set of attributes also appears for a certain set of attributes in another relation.

- A. Logical Integrity
- B. Referential Integrity
- C. Domain Integrity
- D. Data Integrity

ugcnetcse-sep2013-paper3 databases referential-integrity

Answer key 

7.37.1 Relation: UGC NET CSE | July 2018 | Part 2 | Question: 69



Let $R_1(a, b, c)$ and $R_2(x, y, z)$ be two relations in which a is the foreign key of R_1 that refers to the primary key of R_2 . Consider the following four options.

- | | |
|----------------------|----------------------|
| a. Insert into R_1 | b. Insert into R_2 |
| c. Delete from R_1 | d. Delete from R_2 |

Which of the following is correct about the referential integrity constraint with respect to above?

- | | |
|--|--|
| A. Operations a and b will cause violation | B. Operations b and c will cause violation |
| C. Operations c and d will cause violation | D. Operations d and a will cause violation |

ugcnetcse-july2018-paper2 databases relation

Answer key

7.38.1 Relational Algebra: UGC NET CSE | December 2013 | Part 3 | Question: 56



Consider the following schemas:

Branch=(Branch-name, Assets, Branch-city)

Customer = (Customer-name, Bank name, Customer-city)

Borrow = (Branch-name, Loan number, Customer account-number)

Deposit = (Branch-name, Account-number, Customer-name, Balance)

Using relational Algebra, the Query that finds customers who have balance more than 10,000 is _____

- A. $\pi_{customer-name} (\sigma_{balance > 10000} (Deposit))$
 B. $\sigma_{customer-name} (\sigma_{balance > 10000} (Deposit))$
 C. $\pi_{customer-name} (\sigma_{balance > 10000} (Borrow))$
 D. $\sigma_{customer-name} (\pi_{balance > 10000} (Borrow))$

ugcnetcse-dec2013-paper3 databases relational-algebra

Answer key

7.38.2 Relational Algebra: UGC NET CSE | July 2016 | Part 3 | Question: 9



Consider the relations $R(A, B)$ and $S(B, C)$ and the following four relational algebra queries over R and S

- $\pi_{A,B} (R \bowtie S)$
- $R \bowtie \pi_B(S)$
- $R \cap (\pi_A(R) \times \pi_B(S))$
- $\Pi_{A,R,B} (R \times S)$ where $R.B$ refers to the column B in table R

One can determine that:

- | | |
|-------------------------------------|--------------------------------------|
| A. I, III and IV are the same query | B. II, III and IV are the same query |
| C. I, II and IV are the same query | D. I, II and III are the same query |

ugcnetcse-july2016-paper3 databases relational-algebra

Answer key

7.38.3 Relational Algebra: UGC NET CSE | June 2019 | Part 2 | Question: 33



With respect to relational algebra, which of the following operations are included from mathematical set theory?

- Join
- Intersection
- Cartisian product
- Project

A. i and iv

B. ii and iii

C. iii and iv

D. ii and iv

ugcnetcse-june2019-paper2 relational-algebra

Answer key 

7.39

Relational Calculus (2)

7.39.1 Relational Calculus: UGC NET CSE | June 2019 | Part 2 | Question: 31



Which of the following has same expressive power with regard to relational query language?

- i. Rational algebra and domain relational calculus
- ii. Relational algebra and tuples relational calculus
- iii. Relational algebra and domain relational calculus restricted to safe expression
- iv. Relational algebra and tuples relational calculus restricted to safe expression

A. i and ii only

B. iii and iv only

C. i and iii only

D. ii and iv only

ugcnetcse-june2019-paper2 relational-calculus

Answer key 

7.39.2 Relational Calculus: UGC NET CSE | September 2013 | Part 3 | Question: 47



Consider the following schemas:

Branch_Schema = (branch_name, assets, city)

Customer_Schema = (customer_name, street, city)

Deposit_Schema = (branch_name, account_number, customer_name, balance)

Borrow_Schema = (branch_name, loan_number, customer_name, amount)

Which of the following tuple relational calculus finds all customers who have loan amount more than Rs. 12, 0000?

- A. $\{t(\text{customer_name}) \mid t \in \text{borrow}[?] \ t[\text{amount}] > 12000 \}$
- B. $\{t \mid t(\text{customer_name}) \mid t \in \text{borrow}[?] \ t[\text{amount}] > 12000 \}$
- C. $\{t \mid [?] \in \text{borrow} (t(\text{customer_name})=s(\text{customer_name}))[?][\text{amount}]>12000\}$
- D. $\{t \mid [?] \in \text{borrow}(t(\text{customer_name}))[?] \ s[\text{amount}] > 12000\}$

ugcnetcse-sep2013-paper3 databases relational-calculus

7.40

Relational Model (2)

7.40.1 Relational Model: UGC NET CSE | June 2019 | Part 2 | Question: 32



Which of the following features is supported in the rational database model?

A. Complex data-types

B. Multivalued attributes

C. Associations with multiplicities

D. Generalization relationships

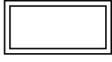
ugcnetcse-june2019-paper2 relational-model

Answer key 

7.40.2 Relational Model: UGC NET CSE | October 2020 | Part 2 | Question: 67



Match List I with List II

List I (E-R symbols)	List II (Description)
(A) 	(I) Key Attribute Type
(B) 	(II) Weak Entity Type
(C) 	(III) Total Participation of Entity in a relation
(D) 	(IV) Multivalued Attribute Type

Choose the correct answer from the options given below:

- A. $A - II, B - IV, C - III, D - I$
 C. $A - II, B - I, C - IV, D - III$

- B. $A - IV, B - I, C - II, D - III$
 D. $A - III, B - IV, C - I, D - II$

ugcnetcse-oct2020-paper2 databases relational-model

Answer key 

7.41

Relational Schema (1)

7.41.1 Relational Schema: UGC NET CSE | November 2017 | Part 3 | Question: 12



Consider a schema $R(A, B, C, D)$ and following functional dependencies.

- $A \rightarrow B$
- $B \rightarrow C$
- $C \rightarrow D$
- $D \rightarrow B$

Then decomposition of R into $R_1(A, B)$, $R_2(B, C)$ and $R_3(B, D)$ is _____

- A. Dependency preserving and lossless join
 B. Lossless join but dependency preserving
 C. Dependency preserving but not lossless join
 D. Not dependency preserving and not lossless join

ugcnetcse-nov2017-paper3 databases relational-schema rdbms

Answer key 

7.42

SQL (33)

7.42.1 SQL: UGC NET CSE | August 2016 | Part 2 | Question: 20



Consider a database table R with attributes A and B . Which of the following SQL queries is illegal ?

- A. $SELECT A FROM R;$
 B. $SELECT A, COUNT(*) FROM R;$
 C. $SELECT A, COUNT(*) FROM R GROUP BY A;$
 D. $SELECT A, B, COUNT(*) FROM R GROUP BY A, B;$

ugcnetcse-aug2016-paper2 databases sql

Answer key 

7.42.2 SQL: UGC NET CSE | August 2016 | Part 3 | Question: 11



Consider the following ORACLE relations :

One $(x, y) = \{ \langle 2, 5 \rangle, \langle 1, 6 \rangle, \langle 1, 6 \rangle, \langle 1, 6 \rangle, \langle 4, 8 \rangle, \langle 4, 8 \rangle \}$

Two $(x, y) = \{ \langle 2, 55 \rangle, \langle 1, 1 \rangle, \langle 4, 4 \rangle, \langle 1, 6 \rangle, \langle 4, 8 \rangle, \langle 4, 8 \rangle, \langle 9, 9 \rangle, \langle 1, 6 \rangle \}$

Consider the following two SQL queries $SQ1$ and $SQ2$:

$SQ1$:

$SELECT * FROM One)$

EXCEPT
 (SELECT * FROM Two);
 SQ2 :
 SELECT * FROM One)
 EXCEPT ALL
 (SELECT * FROM Two);

For each of the SQL queries, what is the cardinality (number of rows) of the result obtained when applied to the instances above ?

- A. 2 and 1 respectively
 C. 2 and 2 respectively
 B. 1 and 2 respectively
 D. 1 and 1 respectively

ugcnetcse-aug2016-paper3 databases sql

Answer key 

7.42.3 SQL: UGC NET CSE | December 2011 | Part 2 | Question: 21



What deletes the entire file except the file structure ?

- A. ERASE
 B. DELETE
 C. ZAP
 D. PACK

ugcnetcse-dec2011-paper2 databases sql

Answer key 

7.42.4 SQL: UGC NET CSE | December 2011 | Part 2 | Question: 22



Which command classes text file, which has been created using "SET ALTERNATIVE" "Command" ?

- A. SET ALTERNATE OFF
 C. CLOSE ALTERNATE
 B. CLOSE DATABASE
 D. CLEAR ALL

ugcnetcse-dec2011-paper2 databases sql

7.42.5 SQL: UGC NET CSE | December 2011 | Part 2 | Question: 24



Which of the following statements is true, when structure of database file with 20 records is modified ?

- A. ? EOF () Prints T
 C. ? BOF () Prints T
 B. ? BOF () Prints F
 D. ? EOF () Prints F

ugcnetcse-dec2011-paper2 databases sql

Answer key 

7.42.6 SQL: UGC NET CSE | December 2011 | Part 2 | Question: 25



The *SQL* Expression Select distinct *T*. branch name from branch *T*, branch *S* where *T*. assets > *S*. assets and *S*. branch-city = DELHI, finds the name of

- A. All branches that have greater asset than any branch located in DELHI.
 B. All branches that have greater assets than allocated in DELHI.
 C. The branch that has the greatest asset in DELHI.
 D. Any branch that has greater asset than any branch located in DELHI.

ugcnetcse-dec2011-paper2 databases sql

Answer key 

7.42.7 SQL: UGC NET CSE | December 2012 | Part 2 | Question: 50



Given a Relation POSITION (Posting-No, Skill), then the query to retrieve all distinct pairs of posting-nos. requiring skill is

- A. Select p.posting-No, p.posting-No from position p where p.skill=p.skill and p.posting-No < p.posting-No

- B. Select p_1 .posting-No, p_2 .posting-No from position p_1 , position p_2 where p_1 .skill= p_2 .skill
- C. Select p_1 .posting-No, p_2 .posting-No from position p_1 , position p_2 where p_1 .skill= p_2 .skill and p_1 .posting.No < p_2 .posting-No
- D. Select p_1 .posting-No, p_2 .posting-No from position p_1 , position p_2 where p_1 .skill= p_2 .skill and p_1 .posting.No = p_2 .posting-No

ugcnetcse-dec2012-paper2 databases sql

Answer key 

7.42.8 SQL: UGC NET CSE | December 2013 | Part 3 | Question: 58



Consider the table student (stuid, name, course, marks). Which one of the following two queries is correct to find the highest marks student in course 5?

Q.1. Select S.stuid from student S where not exist (select * from student e where e course ='5' and e marks $\geq$ s marks)

Q.2. select s.stu.id From student S where s.marks > any (select distinct marks from student S where s.course =5)

- A. Q.1 B. Q.2 C. Both Q.1 and Q.2 D. Neither Q.1 nor Q.2

ugcnetcse-dec2013-paper3 databases sql

Answer key 

7.42.9 SQL: UGC NET CSE | December 2014 | Part 2 | Question: 16



Division operation is ideally suited to handle queries of the type :

- A. Customers who have no account in any of the branches in Delhi.
- B. Customers who have an account at all branches in Delhi.
- C. Customers who have an account in atleast one branch in Delhi.
- D. Customers who have only joint account in any one branch in Delhi

ugcnetcse-dec2014-paper2 databases sql

Answer key 

7.42.10 SQL: UGC NET CSE | December 2014 | Part 2 | Question: 17



Which of the following is true ?

- I. Implementation of self-join is possible in SQL with table alias.
- II. Outer-join operation is basic operation in relational algebra.
- III. Natural join and outer join operations are equivalent.

- A. *I* and *II* are correct. B. *II* and *III* are correct.
- C. Only *III* is correct. D. Only *I* is correct.

ugcnetcse-dec2014-paper2 databases sql

Answer key 

7.42.11 SQL: UGC NET CSE | December 2015 | Part 2 | Question: 13



Consider a "CUSTOMERS" database table having a column "CITY" filled with all the names of Indian cities (in capital letters). The SQL statement that finds all cities that have "GAR" somewhere in its name, is:

A. `select * from customers where city='%GAR%';`

B. `select * from customers where city='$GAR$';`

C. `select * from customers where city like '%GAR%';`

D. `select * from customers where city as '%GAR';`

ugcnetcse-dec2015-paper2 databases sql

Answer key 

7.42.12 SQL: UGC NET CSE | December 2015 | Part 3 | Question: 59



Consider the following database table:

```
Create table test(  
    one integer,  
    two integer,  
    primary key(one),  
    unique(two),  
    check(one>=1 and <=10),  
    check (two>=1 and <=5)  
);
```

How many data records/tuples atmost can this table contain?

- A. 5 B. 10 C. 15 D. 50

ugcnetcse-dec2015-paper3 databases sql

Answer key 

7.42.13 SQL: UGC NET CSE | December 2015 | Part 3 | Question: 60



Suppose ORACLE relation $R(A, B)$ currently has tuples $\{(1, 2), (1, 3), (3, 4)\}$ and relation $S(B, C)$ currently has $\{(2, 5), (4, 6), (7, 8)\}$. Consider the following two SQL queries SQ1 and SQ2:

```
SQ1:     Select * From R Full Join S On R.B=S.B;  
SQ2:     Select * From R Inner Join S On R.B=S.B;
```

The numbers of tuples in the result of the SQL query SQ1 and the SQL query SQ2 are given by:

- A. 2 and 6 respectively B. 6 and 2 respectively
C. 2 and 4 respectively D. 4 and 2 respectively

ugcnetcse-dec2015-paper3 databases sql

Answer key 

7.42.14 SQL: UGC NET CSE | December 2015 | Part 3 | Question: 61



Consider the following three SQL queries (Assume the data in the people table):

- Select Name from people where Age > 21;
- Select Name from people where Height > 180;
- Select Name from people where (Age > 21) or (Height > 180);

If the SQL queries a and b above, return 10 rows and 7 rows in the result set respectively, then what is one possible number of rows returned by the SQL query c ?

- A. 3 B. 7 C. 10 D. 21

ugcnetcse-dec2015-paper3 databases sql

Answer key 

7.42.15 SQL: UGC NET CSE | December 2019 | Part 2 | Question: 22



Given two tables

```
EMPLOYEE (EID, ENAME, DEPTNO)  
DEPARTMENT (DEPTNO, DEPTNAME)
```

Find the most appropriate statement of the given query:

```
Select count (*) 'total'  
from EMPLOYEE  
where DEPTNO IN (D1,D2)
```

```
group by DEPTNO
having count (*) >5
```

- A. Total number of employees in each department $D1$ and $D2$
- B. Total number of employees of department $D1$ and $D2$ if their total is > 5
- C. Display total number of employees in both departments $D1$ and $D2$
- D. The output of the query must have at least two rows

ugcnetcse-dec2019-paper2 sql

Answer key

7.42.16 SQL: UGC NET CSE | January 2017 | Part 2 | Question: 17



Integrity constraints ensure that changes made to the database by authorized users do not result into loss of data consistency. Which of the following statement(s) is (are) true w.r.t. the examples of integrity constraints?

- i. An instructor Id. No. cannot be null, provided Instructor Id No. being primary key.
- ii. No two citizens have same Adhar-Id.
- iii. Budget of a company must be zero.

- A. a, b and c are true.
- B. a false, b and c are true.
- C. a and b are true; c false.
- D. a, b and c are false.

ugcnetjan2017ii databases sql

Answer key

7.42.17 SQL: UGC NET CSE | January 2017 | Part 3 | Question: 11



Consider the following relation:

```
Works (emp_name, company_name, salary)
```

Here, emp name is primary key.

Consider the following SQL query

```
Select emp name
From works T
where salary > (select avg(salary)
                from works S
                where T. company name =
                  S. company name)
```

The above query is for following:

- A. Find the highest paid employee who earns more than the average salary of all employees of his company.
- B. Find the highest paid employee who earns more than the average salary of all the employees of all the companies.
- C. Find all employees who earn more than the average salary of all employees all the companies.
- D. Find all employees who earn more than the average salary of all employees of their company.

ugcnetcse-jan2017-paper3 databases sql

Answer key

7.42.18 SQL: UGC NET CSE | July 2016 | Part 2 | Question: 19



Consider the following two comments C1 and C2 on the relation R from an SQL database:

C1 : drop table R;

C2 : delete from R

- I. Both C1 and C2 delete the schema R
- II. C2 retains relation R, but deletes all tuples in R
- III. C1 deletes not only all tuples, but also the schema for R

- A. I only
- B. I and II only
- C. II and III only
- D. I, II and III only

Answer key

7.42.19 SQL: UGC NET CSE | July 2016 | Part 3 | Question: 11

Consider the following ORACLE relations:

$R(A, B, C) = \{ \langle 1, 2, 3 \rangle, \langle 1, 2, 0 \rangle, \langle 1, 3, 1 \rangle, \langle 6, 2, 3 \rangle, \langle 1, 4, 2 \rangle, \langle 3, 1, 4 \rangle \}$

$S(B, C, D) = \{ \langle 2, 3, 7 \rangle, \langle 1, 4, 5 \rangle, \langle 1, 2, 3 \rangle, \langle 2, 3, 4 \rangle, \langle 3, 1, 4 \rangle \}$

Consider the following two SQL queries:

SQ_1 : SELECT R.B, AVG(S.B) FROM R, S WHERE R.A=S.C AND S.D<7 GROUP BY R.B

SQ_2 : SELECT DISTINCT S.B, MIN(S.C) FROM S GROUP BY S.B HAVING COUNT (DISTINCT S.D)>1;

If M is the number of tuples returned by SQ_1 and If N is the number of tuples returned by SQ_2 then

- A. M=4, N=2 B. M=5, N=3 C. M=2, N=2 D. M=3, N=3

Answer key

7.42.20 SQL: UGC NET CSE | June 2010 | Part 2 | Question: 19

Which data management language component enabled the DBA to define the schema components ?

- A. DML B. Sub-schema DLL C. Schema DLL D. All of these

Answer key

7.42.21 SQL: UGC NET CSE | June 2010 | Part 2 | Question: 20

The PROJECT Command will create new table that has

- A. More fields than the original table B. More rows than original table
C. Both (A) & (B) D. None of these

Answer key

7.42.22 SQL: UGC NET CSE | June 2013 | Part 3 | Question: 8

The employee information of an organization is stored in a relation: Employee (name, sex, salary, deptname). Consider the following SQL query:

```
Select deptname from Employee where sex='M' group by dept name having
avg (salary) > {select avg (salary) from Employee}
```

Output of the given query corresponds to

- A. Average salary of employee more than average salary of the organization
B. Average salary of employee less than average salary of the organization
C. Average salary of employee equal to average salary of the organization
D. Average salary of male employees in a department is more than average salary of the organization

Answer key

7.42.23 SQL: UGC NET CSE | June 2019 | Part 2 | Question: 37

Which of the following statements are DML statements?

- a. Update [tablename] b. Delete [tablename] c. Select * from [tablename]

Set [columnname]=VALUE

- A. a and b B. a and d C. a, b and c D. b and c

ugcnetcse-june2019-paper2 sql

Answer key 

7.42.24 SQL: UGC NET CSE | Junet 2015 | Part 2 | Question: 16



An Assertion is predicate expressing a condition we wish database to always satisfy. The correct syntax for Assertion is:

- A. CREATE ASSERTION 'ASSERTION Name' CHECK 'Predicate'
B. CREATE ASSERTION 'ASSERTION Name'
C. CREATE ASSERTION, CHECK 'Predicate'
D. SELECT ASSERTION

ugcnetcse-june2015-paper2 databases sql

Answer key 

7.42.25 SQL: UGC NET CSE | Junet 2015 | Part 2 | Question: 18



Drop Table cannot be used to drop a Table referenced by _____ constraint

- a. Primary key b. Sub key c. Super key d. Foreign key
A. a B. a, b and c C. d D. a and d

ugcnetcse-june2015-paper2 databases sql

Answer key 

7.42.26 SQL: UGC NET CSE | Junet 2015 | Part 3 | Question: 8



The STUDENT information in a university stored in the relation STUDENT (Name, SEX, Marks, DEPT_Name). Consider the following SQL Query:

```
SELECT DEPT_Name from STUDENT where SEX='M'  
group by DEPT_Name having avg(Marks)>SELECT avg(Marks) from STUDENT
```

It returns the name of the department for which

- A. The average marks of Male students is more than the average marks of students in the department
B. The average marks of Male students is more than the average marks of students in the university
C. The average marks of students is more than the average marks of male students in the university
D. The average marks of Male students is more than the average marks of male students in the university

ugcnetcse-june2015-paper3 sql databases

Answer key 

7.42.27 SQL: UGC NET CSE | November 2017 | Part 3 | Question: 11



Given two relations $R_1(A, B)$ and $R_2(C, D)$, the result of following query

```
Select distinct A, B  
From R1, R2
```

Is guaranteed to be same as R_1 provided one of the following condition is satisfied.

- A. R_1 has no duplicates and R_2 is empty. B. R_1 has no duplicates and R_2 is non-empty.
C. Both R_1 and R_2 have no duplicates. D. R_2 has no duplicates and R_1 is non – empty.

ugcnetcse-nov2017-paper3 databases sql

Answer key 



Consider the following table structures related to a university for the below question.

EMPLOYEE

```
NAME VARCHAR (30) NOT NULL,
EID VARCHAR (10) NOT NULL,
DEPTNO INT (5) NOT NULL,
HODEID VARCHAR (10),
SALARY INT (10),
PRIMARY KEY (EID),
FOREIGN KEY (HODEID) REFERENCES EMPLOYEE (EID),
FOREIGN KEY (DEPTNO) REFERENCES DEPARTMENT (DID);
```

DEPARTMENT

```
DID INT (5) NOT NULL,
DNAME VARCHAR(30) NOT NULL,
HODID VARCHAR (10) NOT NULL,
HODNAME VARCHAR (30),
PRIMARY KEY (DID),
UNIQUE (DNAME),
FOREIGN KEY (HODID) REFERENCES EMPLOYEE (EID)
```

PROJECT WORK:

```
EMPID VARCHAR (10) NOTNULL,
PROJNO INT(5) NOT NULL,
PROJECTLOC VARCHAR (30) NOT NULL,
PRIMARY KEY (EMPID, PROJNO),
FOREIGN KEY (EMPID) REFERENCES EMPLOYEE (EID),
```

Refer table, structures given above, University decided to give all employees in the 'SCIENCE' department a 20% rise in salary. Which of the following query/queries will compute the above results?

- i.

```
UPDATE EMPLOYEE
SET SALARY = SALARY*1.20
WHERE DEPT NO. IN (SELECT DID FROM DEPARTMENT WHERE DNAME = 'SCIENCE');
```
- ii.

```
UPDATE TABLE EMPLOYEE
SET SALARY = SALARY*1.20 WHERE DNAME='SCIENCE';
```
- iii.

```
ALTER TABLE EMPLOYEE
SET SALARY=SALARY*1.20
WHERE DEPTNO. IN (SELECT DNAME FROM DEPARTMENT WHERE DNAME = 'SCEINCE');
```

Choose the correct answer from the options given below:

- A. *a* and *b* only B. *a* only C. *b* and *c* only D. *c* only

ugcnetcse-oct2020-paper2 databases sql

Answer key



Consider the following table structures related to a university for the below question.

EMPLOYEE

```
NAME VARCHAR (30) NOT NULL,
EID VARCHAR (10) NOT NULL,
DEPTNO INT (5) NOT NULL,
```

```

HODEID VARCHAR (10),
SALARY INT (10),
PRIMARY KEY (EID),
FOREIGN KEY (HODEID) REFERENCES EMPLOYEE (EID),
FOREIGN KEY (DEPTNO) REFERENCES DEPARTMENT (DID);

```

DEPARTMENT

```

DID INT (5) NOT NULL,
DNAME VARCHAR(30) NOT NULL,
HODID VARCHAR (10) NOT NULL,
HODNAME VARCHAR (30),
PRIMARY KEY (DID),
UNIQUE (DNAME),
FOREIGN KEY (HODID) REFERENCES EMPLOYEE (EID)

```

PROJECT WORK:

```

EMPID VARCHAR (10) NOTNULL,
PROJNO INT(5) NOT NULL,
PROJECTLOC VARCHAR (30) NOT NULL,
PRIMARY KEY (EMPID, PROJNO),
FOREIGN KEY (EMPID) REFERENCES EMPLOYEE (EID),

```

On the basis of above given table structures, retrieve the distinct employee ID (EMPID) of all employees of university who are working on project. No. 20, 30 and 40

- A. `SELECT EMPID FROM PROJECTWORK WHERE PROJNO=(20,30,40);`
- B. `SELECT EMPID FROM PROJECTWORK WHERE PROJNO IN (20,30,40);`
- C. `SELECT DISTINCT EMPID FROM PROJECTWORK WHERE PROJNO IN(20,30,40);`
- D. `SELECT DISTINCT EMPID FROM PROJECTWORK WHERE PROJNO=20,30,40;`

ugcnetcse-oct2020-paper2 databases sql

Answer key 

7.42.30 SQL: UGC NET CSE | October 2020 | Part 2 | Question: 97



Consider the following table structures related to a university for the below question.

EMPLOYEE

```

NAME VARCHAR (30) NOT NULL,
EID VARCHAR (10) NOT NULL,
DEPTNO INT (5) NOT NULL,
HODEID VARCHAR (10),
SALARY INT (10),
PRIMARY KEY (EID),
FOREIGN KEY (HODEID) REFERENCES EMPLOYEE (EID),
FOREIGN KEY (DEPTNO) REFERENCES DEPARTMENT (DID);

```

DEPARTMENT

```

DID INT (5) NOT NULL,
DNAME VARCHAR(30) NOT NULL,
HODID VARCHAR (10) NOT NULL,
HODNAME VARCHAR (30),
PRIMARY KEY (DID),
UNIQUE (DNAME),
FOREIGN KEY (HODID) REFERENCES EMPLOYEE (EID)

```

PROJECT WORK:

```

EMPID VARCHAR (10) NOTNULL,

```



```

    PROJNO INT(5) NOT NULL,
    PROJECTLOC VARCHAR (30) NOT NULL,
    PRIMARY KEY (EMPID, PROJNO),
    FOREIGN KEY (EMPID) REFERENCES EMPLOYEE (EID),

```

Given below are two statements to find the sum of salaries of all employees of the English department as well as the maximum, minimum and average salary in English department

STATEMENT I:

```

SELECT SUM (SALARY) MAX(SALARY) MIN(SALARY),
AVG (SALARY) FROM EMPLOYEE, DEPARTMENT
WHERE DEPTNO=DID AND DNAME='ENGLISH'

```

STATEMENT II:

```

SELECT SUM (SALARY), MAX(SALARY), MIN (SALARY),
AVG (SALARY), FROM EMPLOYEE, DEPARTMENT
WHERE DNAME='ENGLISH'

```

In the light of the above statements, choose the correct answer from the options given below:

- A. Both Statement I and Statement II are true
- B. Both Statement I and Statement II are false
- C. Statement I is correct but Statement II is false
- D. Statement I is incorrect but Statement II is true

ugcnetcse-oct2020-paper2 databases sql

Answer key 

7.42.31 SQL: UGC NET CSE | October 2020 | Part 2 | Question: 98



Consider the following table structures related to a university for the below question.

EMPLOYEE

```

    NAME VARCHAR (30) NOT NULL,
    EID VARCHAR (10) NOT NULL,
    DEPTNO INT (5) NOT NULL,
    HODEID VARCHAR (10),
    SALARY INT (10),
    PRIMARY KEY (EID),
    FOREIGN KEY (HODEID) REFERENCES EMPLOYEE (EID),
    FOREIGN KEY (DEPTNO) REFERENCES DEPARTMENT (DID);

```

DEPARTMENT

```

    DID INT (5) NOT NULL,
    DNAME VARCHAR(30) NOT NULL,
    HODID VARCHAR (10) NOT NULL,
    HODNAME VARCHAR (30),
    PRIMARY KEY (DID),
    UNIQUE (DNAME),
    FOREIGN KEY (HODID) REFERENCES EMPLOYEE (EID)

```

PROJECT WORK:

```

    EMPID VARCHAR (10) NOTNULL,
    PROJNO INT(5) NOT NULL,
    PROJECTLOC VARCHAR (30) NOT NULL,
    PRIMARY KEY (EMPID, PROJNO),
    FOREIGN KEY (EMPID) REFERENCES EMPLOYEE (EID),

```

Which of the following query/queries return the employee ID and name of employees whose salary is greater than the

salary of all employees in department number 20 of university. Order result by employee ID (refer table structures given above).

- i.

```
SELECT EID, NAME
FROM EMPLOYEE
WHERE SALARY > (SELECT SALARY FROM EMPLOYEE WHERE DEPTNO=20)
ORDER BY EID
```
- ii.

```
SELECT EID, NAME
FROM EMPLOYEE
WHERE SALARY > (SELECT SALARY FROM EMPLOYEE WHERE DEPTNO=20) ;
```
- iii.

```
SELECT EID, NAME
FROM EMPLOYEE
WHERE SALARY > ALL (SELECT SALARY FROM EMPLOYEE WHERE DEPTNO=20)
ORDER BY EID
```

Choose the correct answer from the options given below:

- A. *i* and *ii* only B. *i* and *iii* only C. *ii* only D. *iii* only

ugcnetcse-oct2020-paper2 databases sql

Answer key 

7.42.32 SQL: UGC NET CSE | October 2020 | Part 2 | Question: 99



Consider the following table structures related to a university for the below question.

EMPLOYEE

```
NAME VARCHAR (30) NOT NULL,
EID VARCHAR (10) NOT NULL,
DEPTNO INT (5) NOT NULL,
HODEID VARCHAR (10),
SALARY INT (10),
PRIMARY KEY (EID),
FOREIGN KEY (HODEID) REFERENCES EMPLOYEE (EID),
FOREIGN KEY (DEPTNO) REFERENCES DEPARTMENT (DID);
```

DEPARTMENT

```
DID INT (5) NOT NULL,
DNAME VARCHAR(30) NOT NULL,
HODID VARCHAR (10) NOT NULL,
HODNAME VARCHAR (30),
PRIMARY KEY (DID),
UNIQUE (DNAME),
FOREIGN KEY (HODID) REFERENCES EMPLOYEE (EID)
```

PROJECT WORK:

```
EMPID VARCHAR (10) NOTNULL,
PROJNO INT(5) NOT NULL,
PROJECTLOC VARCHAR (30) NOT NULL,
PRIMARY KEY (EMPID, PROJNO),
FOREIGN KEY (EMPID) REFERENCES EMPLOYEE (EID),
```

In reference to the above given table structures, which of the following query/queries will drop the 'SALARY' column from 'EMPLOYEE' table?

- i.

```
ALTER TABLE EMPLOYEE DROP SALARY CASCADE
```

ii. ALTER TABLE EMPLOYEE DROP SALARY RESTRICT

iii. ALTER EMPLOYEE DROP SALARY

Choose the correct answer from the options given below:

- A. *a* and *b* only B. *a* and *c* only C. *b* and *c* only D. *a* only

ugcnetcse-oct2020-paper2 databases sql

Answer key

7.42.33 SQL: UGC NET CSE | September 2013 | Part 3 | Question: 25



The SQL expression

```
Select distinct T.branch_name from branch T, branch S
where T.assets>S.assets and S.branch_city = "Mumbai"
```

finds the names of

- A. All branches that have greater assets than some branch located in Mumbai
B. All branches that have greater assets than all branches in Mumbai
C. The branch that has greatest asset in Mumbai
D. Any branch that has greater assets than any branch in Mumbai

ugcnetcse-sep2013-paper3 databases sql

Answer key

7.43 Transaction and Concurrency (8)

7.43.1 Transaction and Concurrency: UGC NET CSE | December 2010 | Part 2 | Question: 18



Which of the following is an optimistic concurrency control method ?

- A. Validation based B. Time stamp ordering
C. Lock-based D. None of these

ugcnetcse-dec2010-paper2 databases transaction-and-concurrency

Answer key

7.43.2 Transaction and Concurrency: UGC NET CSE | December 2014 | Part 3 | Question: 08



Which of the following is correct ?

- I. Two phase locking is an optimistic protocol.
II. Two phase locking is pessimistic protocol
III. Time stamping is an optimistic protocol.
IV. Time stamping is pessimistic protocol.

- A. *I* and *III* B. *II* and *IV* C. *I* and *IV* D. *II* and *III*

ugcnetcse-dec2014-paper3 databases transaction-and-concurrency

Answer key

7.43.3 Transaction and Concurrency: UGC NET CSE | January 2017 | Part 3 | Question: 7



Consider following schedules involving two transactions:

$S_1 : r_1(X); r_1(Y); r_2(X); r_2(Y); w_2(Y); w_1(X)$

$S_2 : r_1(X); r_2(X); r_2(Y); w_2(Y); r_1(Y); w_1(X)$

Which of the following statement is true?

- A. Both S_1 and S_2 are conflict serializable.
- B. S_1 is conflict serializable and S_2 is not conflict serializable
- C. S_1 is not conflict serializable and S_2 is conflict serializable
- D. Both S_1 and S_2 are not conflict serializable

ugcnetcse-jan2017-paper3 databases transaction-and-concurrency

Answer key 

7.43.4 Transaction and Concurrency: UGC NET CSE | June 2012 | Part 2 | Question: 12



A Transaction Manager is which of the following?

- A. Maintains a log of transactions
- B. Maintains before and after database images
- C. Maintains appropriate concurrent control
- D. All of the above

ugcnetcse-june2012-paper2 databases transaction-and-concurrency

Answer key 

7.43.5 Transaction and Concurrency: UGC NET CSE | November 2017 | Part 3 | Question: 7



Consider the following four schedules due to three transactions (indicated by the subscript) using read and write on a data item X , denoted by $r(X)$ and $w(X)$ respectively. Which one of them is conflict serializable?

$S_1: r_1(X); r_2(X); w_1(X); r_3(X); w_2(X)$

$S_2: r_2(X); r_1(X); w_2(X); r_3(X); w_1(X)$

$S_3: r_3(X); r_2(X); r_1(X); w_2(X); w_1(X)$

$S_4: r_2(X); w_2(X); r_3(X); r_1(X); w_1(X)$

- A. S_1
- B. S_2
- C. S_3
- D. S_4

ugcnetcse-nov2017-paper3 databases transaction-and-concurrency

Answer key 

7.43.6 Transaction and Concurrency: UGC NET CSE | November 2017 | Part 3 | Question: 8



Suppose a database schedule S involves transactions $T_1, T_2, \dots, T_n$. Consider the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?

- A. Topological order
- B. Depth – first order
- C. Breadth – first order
- D. Ascending order of transactions indices

ugcnetcse-nov2017-paper3 databases transaction-and-concurrency

Answer key 

7.43.7 Transaction and Concurrency: UGC NET CSE | September 2013 | Part 2 | Question: 37



Usage of Preemption and Transaction Rollback prevents ____.

- A. Unauthorized usage of data file
- B. Deadlock situation
- C. Data manipulation
- D. File pre-emption

ugcnetsep2013iii databases transaction-and-concurrency

Answer key 

7.43.8 Transaction and Concurrency: UGC NET CSE | September 2013 | Part 2 | Question: 43



Thoma's-write rule is

- A. Two phase locking protocol
- B. Timestamp ordering protocol
- C. One phase locking protocol
- D. Sliding window protocol

ugcnetsep2013iii databases transaction-and-concurrency

Answer key 

7.44

Transactions and Concurrency Control (2)

7.44.1 Transactions and Concurrency Control: UGC NET CSE | June 2023 | Part 2: 44



Which one of the following is NOT a part of ACID properties of a database transaction?

- A. Atomicity B. Consistency C. Isolation D. Deadlock-freedom

ugcnetcse-june2023-paper2 databases transactions-and-concurrency-control

Answer key

7.44.2 Transactions and Concurrency Control: UGC NET CSE | October 2020 | Part 2 | Question: 48



In the context of concurrency control, a given pair of operations in a schedule is called conflict schedule if

- At least one of the operations is write operation
- Both the operations are performed on the same data item
- Both the operations are performed by different transactions
- Both the operations are performed on different data items

Choose the correct answer from the options given below:

- A. (a) and (b) only B. (a), (b) and (c) only
C. (a), (c) and (d) only D. (c) and (d) only

ugcnetcse-oct2020-paper2 databases transactions-and-concurrency-control

Answer key

7.45

Weak Entity (1)

7.45.1 Weak Entity: UGC NET CSE | Junet 2015 | Part 2 | Question: 20



For a weak entity set to be meaningful, it must be associated with another entity set in combination with some of their attribute values, is called as:

- A. Neighbour Set B. Strong Entity Set
C. Owner Entity Set D. Weak Set

ugcnetcse-june2015-paper2 databases weak-entity

Answer key

Answer Keys

7.0.1	C	7.0.2	D	7.0.3	D	7.0.4	C	7.0.5	C
7.0.6	B	7.0.7	B	7.0.8	D	7.0.9	B	7.0.10	D
7.0.11	A	7.0.12	D	7.0.13	D	7.0.14	Q-Q	7.0.15	Q-Q
7.0.16	Q-Q	7.0.17	Q-Q	7.0.18	Q-Q	7.0.19	Q-Q	7.0.20	Q-Q
7.0.21	Q-Q	7.0.22	Q-Q	7.0.23	Q-Q	7.0.24	Q-Q	7.0.25	Q-Q
7.0.26	Q-Q	7.0.27	Q-Q	7.0.28	Q-Q	7.0.29	Q-Q	7.1.1	Q-Q
7.1.2	C	7.2.1	A	7.2.2	A	7.3.1	X	7.4.1	Q-Q
7.5.1	B	7.5.2	C	7.5.3	B	7.5.4	C	7.6.1	B
7.7.1	A	7.8.1	Q-Q	7.9.1	Q-Q	7.10.1	B	7.11.1	Q-Q
7.11.2	D	7.12.1	B	7.13.1	Q-Q	7.13.2	A	7.14.1	D
7.14.2	A	7.15.1	A	7.16.1	Q-Q	7.16.2	B	7.17.1	N/A
7.17.2	D	7.17.3	A	7.17.4	B	7.17.5	Q-Q	7.17.6	A

7.17.7	B
7.17.12	A
7.17.17	A
7.17.22	C
7.19.1	A
7.21.3	Q-Q
7.21.8	C
7.21.13	A
7.21.18	C
7.26.1	B
7.30.2	Q-Q
7.34.1	C
7.34.6	D
7.36.2	B
7.38.3	B
7.41.1	A
7.42.5	Q-Q
7.42.10	D
7.42.15	Q-Q
7.42.20	Q-Q
7.42.25	C
7.42.30	C
7.43.2	D
7.43.7	B

7.17.8	D
7.17.13	D
7.17.18	B
7.17.23	Q-Q
7.20.1	Q-Q
7.21.4	Q-Q
7.21.9	D
7.21.14	Q-Q
7.22.1	B
7.27.1	Q-Q
7.31.1	B
7.34.2	A
7.34.7	C
7.36.3	B
7.39.1	B
7.42.1	B
7.42.6	Q-Q
7.42.11	C
7.42.16	C
7.42.21	Q-Q
7.42.26	B
7.42.31	D
7.43.3	C
7.43.8	B

7.17.9	B
7.17.14	A
7.17.19	C
7.18.1	C
7.20.2	B
7.21.5	Q-Q
7.21.10	A
7.21.15	B
7.23.1	D
7.28.1	A
7.31.2	C
7.34.3	Q-Q
7.34.8	A
7.37.1	D
7.39.2	C
7.42.2	B
7.42.7	C
7.42.12	B
7.42.17	D
7.42.22	D
7.42.27	B
7.42.32	A
7.43.4	D
7.44.1	Q-Q

7.17.10	B
7.17.15	Q-Q
7.17.20	D
7.18.2	A
7.21.1	Q-Q
7.21.6	A
7.21.11	Q-Q
7.21.16	B
7.24.1	B
7.29.1	B
7.32.1	B
7.34.4	Q-Q
7.35.1	B
7.38.1	A
7.40.1	C
7.42.3	Q-Q
7.42.8	B
7.42.13	B
7.42.18	C
7.42.23	C
7.42.28	B
7.42.33	A
7.43.5	D
7.44.2	B

7.17.11	B
7.17.16	Q-Q
7.17.21	A
7.18.3	C
7.21.2	Q-Q
7.21.7	B
7.21.12	A
7.21.17	A
7.25.1	Q-Q
7.30.1	B
7.33.1	Q-Q
7.34.5	A
7.36.1	B
7.38.2	D
7.40.2	C
7.42.4	Q-Q
7.42.9	B
7.42.14	C
7.42.19	A
7.42.24	A
7.42.29	C
7.43.1	Q-Q
7.43.6	A
7.45.1	C